# FACTORS THAT AFFECT ONLINE CUSTOMER SATISFACTION TOWARDS GOODS DELIVERY SERVICES IN KLANG VALLEY

By

**TEE JU EN**

**VENUS SOONG FANG YING**

A project report submitted to the Department of Physical and Mathematical Science

Faculty of Science

Universiti Tunku Abdul Rahman

in partial fulfilment of the requirements for the degree of

Bachelor of Science (Hons) Logistics and International Shipping

March 2020

## ABSTRACT

**FACTORS THAT AFFECT ONLINE CUSTOMER SATISFACTION TOWARDS GOODS DELIVERY SERVICES IN KLANG VALLEY**

**TEE JU EN**

**VENUS SOONG FANG YING**

With the quick development of online shopping, delivery service provider has become extremely significant in development of goods within the supply chain network. In recent years, online shopping becomes a growing trend in Malaysia. Many people tend to purchase goods online instead of going to the physical store as it provide convenience to online consumer in skipping the traffic and long queues. However, the expanding of delivery services unable to perform as customer expectation sometimes. The objectives of this research is to identify and examine the criteria and sub criteria of goods delivery services that can influence consumers' online shopping satisfaction among people who purchase things online in Klang Valley. Primary data were gathered through study survey with 200 respondents who have experience in buying goods from online store. In addition, this research also determines the most preferred factor among decision criteria and sub criteria by using AHP model. The result of this project shows reliability which ranked the first is the main factor that can affect consumer satisfaction in goods delivery services in Klang Valley. At last, this study will discuss the limitations and recommendations of this research. Keywords: Online Shopping, Delivery Services, Customer Satisfaction, Analytic Hierarchy Process

## ACKNOWLEDGEMENT

First of all, we would like to take this opportunity to thank our research supervisor, Dr. Lam Weng Hoe who have been supporting, guiding and assisting us to complete this research project. Without the continuous assistance with his patience and encouragement, we will not have the confidence to complete this research project. He has leaded us in getting ideas, conducting survey and performing analysis. Moreover, we would like to thank all of the lecturers and tutors who had been teaching us and provide us many knowledge. We also feel grateful to all the respondents who had participated in our questionnaire survey in order to help our research successfully.

Furthermore, we want to show our gratitude to our families who always been there support us and do not give us pressure so that we can perform well in our studies. Last but not least, we appreciation our friends and course mates who always motivate and keep track of the progress each other. As a result, we are able to complete our research on time. None of this could be happened without the supporting of these people in our university study life.

## DECLARATION

We hereby declare that this research project is the end result of our original work except for the duly acknowledged which all the citations have been given in the references. We also declare that this research project has not been submitted for any other degree qualification in UTAR or other institutes of learning.

_____

**TEE JU EN**

_____

**VENUS SOONG FANG YING**

## APPROVAL SHEET

This final year project report entitled "**FACTORS THAT AFFECT ONLINE CUSTOMER SATISFACTION TOWARDS GOODS DELIVERY SERVICES IN KLANG VALLEY**" was prepared by TEE JU EN and VENUS SOONG FANG YING. This project was submitted as partial fulfillment of the requirements for the degree of Bachelor of Science (Hons) Logistics and International Shipping at Universiti Tunku Abdul Rahman.

Approved by:

_____

Dr . Lam Weng Hoe Date:……..
Supervisor
Department of Physical and Mathematical Science
Faculty of Science
Universiti Tunku Abdul Rahman

**FACULTY OF SCIENCE UNIVERSITI TUNKU ABDUL RAHMAN**

**Date: _____**

**PERMISSION SHEET**

It is hereby certified that **TEE JU EN** (ID No: **17ADB01215**) and **VENUS SOONG FANG YING** (ID No: **17ADB01820**) have completed this final year project report entitled "FACTORS THAT AFFECT ONLINE CUSTOMER SATISFACTION TOWARDS GOODS DELIVERY SERVICES IN KLANG VALLEY" under the supervision of DR. LAM WENG HOE from the Department of Physical and Mathematical Science, Faculty of Science.

I hereby give permission to the University to upload the softcopy of my final year project report in pdf format into the UTAR Institutional Repository, which may be made accessible to the UTAR community and public.

Yours truly,

_____

(TEE JU EN)

_____

(VENUS SOONG FANG YING)

## TABLE OF CONTENTS


| | Page |
|---|---|
| **ABSTRACT** | **ii** |
| **ACKNOWLEDGEMENTS** | **iii** |
| **DECLARATION** | **iv** |
| **APPROVAL SHEET** | **v** |
| **PERMISSION SHEET** | **vi** |
| **TABLE OF CONTENTS** | **vii** |
| **LIST OF TABLES** | **xii** |
| **LIST OF FIGURES** | **xiii** |
| **LIST OF ABBREVIATIONS** | **xiv** |

**CHAPTER**

1. **INTRODUCTION**

| | |
|---|---|
| 1.0. Introduction | 1 |
| 1.1. Background of Study | 3 |
| 1.2. Problem Statement | 7 |
| 1.3. Research Objectives | |
| 1.3.1 Specific Objectives | 10 |
| 1.3.2 Research Questions | 11 |
| 1.4. Significance of the Study | 11 |
| 1.5. Scope and limitation of Study | 12 |
| 1.6. Chapter Layout | 13 |
| 1.7. Conclusion | 14 |

## 2. LITERATURE REVIEW


2.0 Introduction 15

2.1 Criteria and Sub Criteria

2.1.1 Timeliness

2.1.1.1 Lead Time Management 16

2.1.1.2 Receiving and Delivering Time 16

2.1.1.3 Just-in Time System 18

2.1.2 Reliability

2.1.2.1 Accuracy of Online Order Service 19

2.1.2.2 Availability and Security of Website 20

2.1.2.3 Accurate Delivery Service 20

2.1.3 Responsiveness

2.1.3.1 Problem Solving 22

2.1.2.2 Interaction between Employees and Customer 22

2.1.2.3 Access to Information 23

2.1.4 Assurance

2.1.4.1 Quality of Personnel 24

2.1.5 Delivery cost

2.1.5.1 Competitive Advantage with Lower Cost 26

2.1.5.2 Free Delivery 27

2.2 Customer Satisfaction 28

2.3 Purposed Theoretical Framework 32

**3. METHODOLOGY**

3.0 Introduction 33

3.1 Research Design 33

3.2 Data Collection

3.2.1 Primary Data 34

3.3 Sampling Design

3.3.1 Target Population 36

3.3.2 Sampling Frame and Location 36

3.3.3 Sampling Unit 37

3.3.4 Sampling Elements 37

3.3.5 Sampling Size 37

3.3.6 Sampling Method 38

3.4 Model Development 39

3.5 Research Instrument

3.5.1 Questionnaires Design 41

3.6 Data Analysis Procedures 42

3.7 Conclusion 47

**4. RESULTS AND DISCUSSION**

4.0 Introduction 48

4.1 Pairwise Comparison Matrices

4.1.1 Comparison among Decision Criteria 49

4.1.2 Comparison among sub criteria- Timeliness 52

4.1.3 Comparison among sub criteria- Reliability 54

4.1.4 Comparison among sub criteria- Responsiveness 57

4.1.5 Comparison among sub criteria- Assurance 59

4.1.6 Comparison among sub criteria- Delivery Cost 59

4.2 Result Analysis 61

4.3 Consistency Ratio 63

4.4 Summary of Findings 65

4.5 Recommendations 66

4.6 Contribution of the Research Project 67

**5. CONCLUSION**

5.0 Introduction 68

5.1 List for the Research Goals and Study Questions

5.1.1 Research Questions One 69

5.1.2 Research Questions Two 70

5.1.3 Research Questions Three 71

5.2 Implications for Theory

5.2.1 Implications for Delivery Services Providers 72

5.3 Limitation of Study 73

5.4 Future Research of Study 74

5.5 Conclusion 74

**REFERENCES** 76

**APPENDICES**

Appendix A- Questionnaire 81

## LIST OF TABLES


| Table | Page |
|---|---|
| 3.1 Hierarchy level of Consumer Satisfaction towards distribution services | 40 |
| 3.2 Saaty Scale | 43 |
| 3.3 Random Consistency Index | 46 |
| 4.1: Pair-wise comparison matrix for all five decision criteria | 49 |
| 4.2: Normalization of five decision criteria | 50 |
| 4.3: Weight score and ranking of five decision criteria | 51 |
| 4.4 Comparison among sub criteria- Timeliness | 52 |
| 4.5: Normalization of sub criteria- Timeliness | 53 |
| 4.6: Weight score and ranking of sub criteria- Timeliness | 53 |
| 4.7 Comparison among sub criteria- Reliability | 55 |
| 4.8: Normalization of sub criteria- Reliability | 55 |
| 4.9: Weight score and ranking of sub criteria- Reliability | 56 |
| 4.10 Comparison among sub criteria- Responsiveness | 57 |
| 4.11: Normalization of sub criteria- Responsiveness | 58 |
| 4.12: Weight score and ranking of sub criteria- Responsiveness | 58 |
| 4.13 Comparison among sub criteria- Delivery Cost | 59 |
| 4.14: Normalization of sub criteria- Delivery Cost | 60 |
| 4.15: Weight score and ranking of sub criteria- Delivery Cost | 60 |

## LIST OF FIGURES


| Table | Page |
|---|---|
| 1.1: Number of courier licenses in Malaysia | 4 |
| 1.2 Postal Service and Courier Service | 6 |
| 3.1 AHP index | 39 |
| 4.1: Weight of five decision criteria | 51 |
| 4.2: Results on timeliness factor | 54 |
| 4.3: Results on reliability factor | 56 |
| 4.4: Results on responsiveness factor | 58 |
| 4.5: Results on delivery cost factor | 60 |

## LIST OF ABBREVIATIONS

| | |
|---|---|
| CPL | Courier Partner Logistic |
| GST | Good and Service Tax |
| SST | Sale and Service Tax |
| IoT | Internet of Things |
| VMI | Vendor Management Inventory |
| JIT-L | Just-In-Time Logistics |
| RFID | Radio Frequency Identification |
| AHP | Analytic Hierarchy Process |
| MCDM | Multiple Criteria Decision Making |
| WSV | Weighted Sum Vector |
| CV | Consistency Vector |
| CI | Consistency Index |
| CR | Consistency Ratio |
| RI | Random Index |
| λmax | Maximum eigenvalue |
| n | Number of decision criteria |

# CHAPTER 1

# INTRODUCTION

## 1.0 Introduction

In this chapter 1, we will discuss the origin of this exploration, the problem statement, research objectives and also research questions. Then followed by the significance of the studies, the scope and limitation and also chapter layout. A brief description about this chapter will be given toward the finish of this part.

In our nation - Malaysia has enormous capacity for the development of the demand of e-commerce, as well as the nation seen improved internet connectivity, strong smart phone usage and enhanced security. The third most noteworthy nation in Southeast Asia in view of usage on the internet is Malaysia. The e-commerce sector in our nation has expanded at 31 per cent between 2010 and 2014 and the business size has risen relied upon to achieve at USD 3.1 billion until 2020. (Fadilah Siali, 2018) E-commerce is changing the courier and delivery service in Malaysia. The fast development of e-commerce is positive for courier and delivery service providers and also new players in the business. Due to critical privatization, globalization and advancement of the e-commerce, the delivery services business are developing rapidly and become a growing service industry with competitive prices and flexible services. (Otsetova, 2017)

With the high demand for e-business deliveries, there would be new dispatch organizations in Malaysia, described by high quantities of little volume shipments. Players in the delivery service industry would look into associations and consolidation to extend their system. Top logistic players also hope to grow into the fast-developing delivery service business to expand the players' income streams. To get rid throughout the world of today, courier associations have to improve by deregulation in order to turn out to be more aggressive, market-based and consumer-driven. Throughout this case, the costs are no longer the primary adjective of the courier business. Responsiveness like pace and efficiency will be the next source of competitive advantage particularly in the industry of courier business. (Alminnourliza Noordin, n.d.)

The efficiency of the courier service has contributed to more consumer trust and satisfaction. (Jessica Sze Yin Ho, 2012) Customer satisfaction is becoming more and more significant for organizations and it has an immediate effect on the organization's benefit. Online purchasers expect higher levels of delivery service quality contrast with traditional channels customers. (KEVIN CHAI YEE SHENG, 2018) At the point when expectations match the performance, it is said that clients are satisfied, while the clients are disappointed when expectations are lower than the performance. Highly satisfied consumers remain faithful to the organization for a longer period. They will buy more when the organization presents new products or services. (Piruni A. Deyalage, 2019) Moreover, they tend to shop back again which the company can gain competitive advantages.

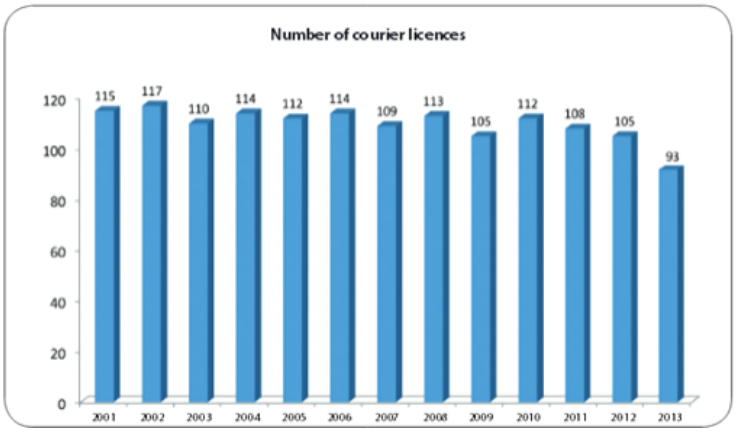
## 1.1 Background of Study

The dispatch business has progressed a long way after the time of horseback distribution, it is actually being operated by specialist carriers and along with multiple resellers with comprehensive network extend all across the globe. Advancing shipping has ensured that packages can now be transported across the globe only in current day. Advancements in innovation and the web have opened up worldwide package delivery to each one and have allowed huge enhancements to the services offered by the greatest players in the business.

Throughout the years, as economies create and new advances develop, delivery services have experienced progressive changes. Postal administrators have taken up new administrations while divesting others. (Lee, 2001) Since the late 1990s onwards, Malaysia looks ahead to be an industry powered by development. Some of the biggest contributions to this accomplishment has been rendered by the service sector. Courier business is a component of the service industry that continues to expand. In 2008, global mail and parcel distribution was projected to cross USD500 billion and is expected to keep expanding. (Alminnourliza Noordin, n.d.) The distribution services sector in Malaysia has been experiencing growing competition since the country's first international service provider, DHL, joined the 1970s. (Jessica Sze Yin Ho, 2012)

The delivery service sector advances toward a focused industry in Malaysia then to competition among expedited delivery specialists in nation is extraordinary. Both of the local and worldwide organizations are intensively assume in Malaysia and any company is seeking to earn further market share from the business as a whole. To increase their competitive advantage, numerous organizations are attempting to deal with their logistic activities and that is the explanation for the admission of third party logistics to Malaysia. Ones have often rendered use of their programs. In country Malaysia, clients utilize logistics by third parties to carry their products all around the globe. All third party operations are expertise in cargo sending, container haulage and warehouse activity. (Naser Valaei, 2016) Malaysian Communication and Multimedia Commission make know that 115 transport licenses had approved in the year 2001. (YUSSOFF, 2014)

**Figure 1.1:** Number of courier licenses in Malaysia




It is basic in Malaysia in logistics services like delivery services organizations which they call clients for whom purchase their services and for e-commerce business that they call delivery services organizations as a colleague or Courier Partner Logistics (CPL). (Nurul Izzah, 2016)

There are a range of logistics firms providing parcel distribution services in Malaysia, the biggest is Pos Malaysia Berhad which is built up in 1988. This organization owns the biggest dispatch team and the largest network infrastructure in the nation. (Fadilah Siali, 2018) At that time, letters were sent on to dispatchers or to specific distribution men. Instead of postage stamps, fines were charged while letters had been delivered to the post office. A certificate was then issued to the letters sent.

Generally, postal and courier business are known with the distribution of orders, packages, documents, letters and written materials. Delivery process involves the processing of mail, input arranging, composing the creation of postal systems, yield arranging and distribute post. Due to the difficulty of the procedure, ICT will be supported to guarantee the appropriate post will be obtained by the client just on schedule. (Alminnourliza Noordin, n.d.) In the past courier, services were constrained distinctly to delivery of physical communication documents, for example parcel and post delivery. The dispatch business is anyway advanced into the delivery of current mail facilities, delivery facilities, freight facilities and online retails for their clients in current years. (Bishop Emmanuel Prince, 2016)

**Figure 1.2** Postal Service and Courier Service

| Postal Service | |
|---|---|
| Postal services related to letters | Services consisting of pick-up, transport and delivery services of letters, newspapers, journals, periodicals, brochures, leaflets and similar printed matters, whether for domestic or foreign destinations, as rendered by the national postal administration. |
| Postal services related to parcels | Services consisting of pick-up, transport and delivery services of parcels and packages, whether for domestic or foreign destinations, as rendered by the national postal administration. |
| Post office counter services | Services rendered at post office counters, e.g. sales of postage stamps, handling of certified or registered letters and packets, and other post office counter services. |
| Other postal services | Mailbox rental services, "poste restante" services, and public postal services not elsewhere classified. |
| **Courier Service** | |
| Multi modal courier services | Services consisting of pick-up, transport and delivery services, whether for domestic or foreign destinations of letters, parcels and packages, rendered by courier and using one or more modes of transport, other than by the national postal administration. These services can be provided by using either self-owned or public transport media. |
| Other courier services | Other courier services for goods, not elsewhere classified, e.g./trucking or transfer services without storage, for freight. |

The way to achievement in the present demanding competitive environment is controlled by the delivery of quality services and this will enhance the degree of consumer loyalty and client standard for dependability. The issue of consumer loyalty and service quality is crucial for research as associations attempt to improve their service quality for consumer satisfaction and loyalty. (S. Akilimalissiga, 2017) Even though service quality positively affects the organization, it is not a guarantee of changing over current purchasers into long haul faithful clients, nor does it normally protect the organization from clients changing to options. Consequently, an affiliation partnership must be formed in order to gain client loyalty. Faith is the main element in the partnership between buyer and dealer to develop continuously. (Chou, 2014)

## 1.2 Problem Statement

The expanding number of parcels has made delivery Service Company unable to perform as indicated by their guarantee, for example for more often, the package was not delivered in time. This had caused client disappointment to the Regional Courier Systems and they may change to a specific courier business firm. (Fadilah Siali, 2018) In the e-commerce world, delivery time plays a significant role in the fulfillment and management of clients. Timing has a major impact on the partnership between the content of the online store and the sales. (Zulkarnain Kedah, 2015) Additionally, this industry faces difficulties because of globalization. Although Malaysia's delivery sector expanded at 7% each year, and the business has numerous troubles ahead in terms of low overall sales, exposure to finance, receipt of technologies and productivity. (Alminnourliza Noordin, n.d.)

Meanwhile, another case showed that an organization's responsiveness towards the client is very important. For instance, the package had damage and loss during the delivery process. The clients had submitted a complaint, however National Courier Services fails to commit for the harm and costs. As a dispatch organization, the National Courier Services Provider will offer timely service to its clients and be liable for the distribution of the order. Global Courier Companies provided a significant number of concerns over sometimes late refreshment of their package distribution monitoring data, rendering it difficult for clients to monitor the path of their package. (Fadilah Siali, 2018) As a delivery service organization, refreshing the following records is significant as this will offer confirmation to the client about their cargo delivery.

Furthermore, another client objection was made that his partner was having difficulty walking, unable to receive the package at the mail processing centre of the company, even though he delivered an approval statement to receive package. Even though the plaintiff acknowledged that it had been because the package was MasterCard and the issuer of the credit card would show for processing, he claimed that he and his partner were stay in their house in the loft unit on the actual day of distribution, sitting tight to wait the package. Nevertheless, they did not get a call from the postal worker or the security officer regarding the distribution. Likewise, the notice of the attempted distribution did not indicate the date. This incident shows that empathy affects consumer loyalty. (Fadilah Siali, 2018)

Another component that may influence consumer loyalty is cost. Since the execution of Sale and Service Tax (SST) in September 2018, all merchandise and ventures as well as for postal and shipping facilities, six per cent were taxable. Even though the tax is charged to the state, but is received by supplier. (Fadilah Siali, 2018) This has increased the price of postage and delivery service. Postage fees are no longer a small cost.

Organizations that provide delivery service are famously battling with similar issues such as fuel costs development and road infrastructure. Numerous delivery service organizations' impacts of mistakes of the evaluating pricing strategy, made during ongoing years turn up. These organizations continually inaccurately brought down costs of their services, rather than

attempting to keep them, taking into consideration even the degree of market inflation. (Jacek Karcz, 2016) The primary impacts of high fuel costs can be seen inside the macro-economic framework. The principal effect reflects the role of transportation in deciding the cost of cargo. Higher fuel prices can also cause a higher delivery expense.

Client service considerations just as distribution cost factors play a position in selecting a service offering. Providing additional time frames can expand client support yet it's going to decrease fall intensity for example the quantity of trips the mail van makes in a specific postal district which, in effect, which lead to increased shipping costs. Besides, shorter period can offer more convenience to the customer. However, route versatility may be decreased and distribution costs may be increased. (Niels Agatz, n.d.)

The local financial trend represents the rapid development of the financial system for the last several years. In Malaysia, this auxiliary transformation had offered a more robust approach that showed the growth of large-scale ecosystems. As the world's big-scale ecosystem was changing slowly, the mail business was met with the increasing usage of emerging technologies. As a consequence of the advanced boom, competition for conventional postal services is increasing. In 2013, Pos Malaysia announced that Ringgit Malaysia's revenue of 1.27 billion was decided by transport and retail sector. Distribution and logistics company has seen a twofold rise in competition, contract companies and rapid growth of the package.

This research will likewise concentrate on delivery service providers' creativity business model, investigations the issues of quality progression and recommendations for strengthening consumer loyalty problems. As a consequence, the five targeted courier sector would have a stable competitive edge. Strong quality creativity should be done straightforwardly impact the interaction with consumers, knowledge, capabilities and enhanced self-help, eliminating holding time by allowing whole day and 7 days of entry utilizing innovation gadgets like internet browsers, cell phones and kiosks. (YUSSOFF, 2014)

## 1.3 Research Objectives

The purpose of this thesis was to analyze the aspects that influence online consumer loyalty and disappointment with courier services in Klang Valley and discuss methods to boost customer experience in delivery services.

### 1.3.1 Specific Objectives

i. To understand the factors that affect online customer satisfaction towards delivery services
ii. To identify the ranking of factors that affect online customer satisfaction towards delivery services by using the AHP model

iii. To determine the most preferable factors among decision criteria based on the online customer in Klang Valley

### 1.3.2 Research Questions

i. What are the factors that affect online customer satisfaction towards delivery services?

ii. How online customers in Klang Valley ranked their satisfaction towards delivery service quality in terms of the factors?

iii. Which factor of delivery services is more preferred for online customers?

## 1.4 Significance of the Study

In this rapidly developing world, saving time has a basic impact on our regular daily life. A few examinations have been carrying out at the point of internet retail and its connection to the distribution service in order to decide the most ideal method an individual can fulfill their needs. (Cihan, 2017) Customers believe poor delivery standards to be the most noticeably awful thing that can turn out badly with internet shopping. Most consumers use e-commerce services for ease, but if deliveries are late or mistaken it might avert them coming back to purchase more, harming trust and affecting reputation. The purpose that doing this report is just to recognize the consider variables that have an impact on online consumer service towards delivery services since this process plays an significant part in our daily lives. This can directly affect consumer behavior in

online shopping. Besides, this thesis may render a relevant link to investigators. Through this study, investigators have a better understanding and knowledge of customer satisfaction towards delivery services. In future work, investigators may use certain forms of techniques to analyze the variables concerned which can influence online client satisfaction and dissatisfaction in delivery services.

### 1.5 Scope and Limitation of Study

The key aim of the whole analysis will be to understand the factors influencing online client loyalty towards delivery services. Around the same period, to analyze whether sub criteria such as delivery time, the consistency of the request, the accuracy of the customer's information, the ability of employees to deliver trust and distribution prices have a huge effect on consumer loyalty in the distribution phase in Klang Valley.

This examination is restricted under a few limitations which recommend the avenues for work in the future. Firstly, this kind of work was more focused on the perception of respondents due to time and budget constraints. Future research utilizing a more random strategy would give a better representation of tests. Besides, using citizens in Klang Valley as the unit of analysis restricted the predictive validity of the whole investigation. Examination of survey participants from

various categories of respondents includes employed parents and companies may give a specific variety of service level desires across third parties.

## 1.6 Chapter Layout

Chapter 1: Introduction

This segment provides the understanding of the interaction among the factors, in respect on prior analysis and statistics. There are six subchapters to outline the research.

Chapter 2: Literature review

The highlighted factors and a theoretical framework for the comparison of the related studies will be provided in this segment.

Chapter 3: Research Methodology

Each chapter provides data collection methods focused on the applicable analysis design and data collection process used to identify the source of the results. The target audience is also defined by sample nature, which is split into six subchapters to be analyzed. The chapter begins with model development. The info is also combined with research instrument as well as data analysis procedures. This review concluded with a chapter summarizing the data collected to increase the legitimacy of the research proposal.

## 1.7 Conclusion

The whole section offers a description of the whole research study and explains the context of the supply chain in e-commerce, the questions relevant to research issues, priorities, significance of this project and limitation of research. Then next chapter addressed previous study projects and defined future goals and introduced the theoretical structure implemented of this research paper.

# CHAPTER 2

# LITERATURE REVIEW

## 2.0 Introduction

This section is a critical evaluation of sources related to the research topic. It involves analyzing and synthesizing past studies that had discussed by previous researchers. All the factors will be featured and explained in this part. Many theories have been proposed to explain which factors could affect the satisfaction of delivery services among online customers in Klang Valley. There have a few articles that had investigated the topic of doing customers satisfy with current delivery service and delivery service quality that could affect consumers' satisfaction. Timeliness is indeed one of the reasons which influence client loyalty because clients are being impatient and unsatisfied when they must be postponed or wait for longer occasions to wait for their parcel to come. (Okyere, et al., 2015) Besides, reliability can also affect customer satisfaction as accurate and complete order service has a significant influence on increase satisfaction in return encouraging future repurchases. (Omar, et al., 2015) Moreover, responsiveness is another reason that could make dissatisfaction in customers since it improves the efficiency of delivery service. (Chhetri, et al., 2017) In addition, assurance which means of being safe also the reason customer feel confident toward the delivery service. Customers are more satisfy and enjoy purchasing products and services with a low delivery costs. (Usmana & Rehman, 2017)

## 2.1 Criteria and Sub Criteria

### 2.1.1 Timeliness

#### 2.1.1.1 Lead Time Management

Lead timeframe is indeed the period limit for the vendor to supply the contracted items to the client or carrier as stated in the contract of sale. The vendor must deliver specific goods on the deadline or period stated throughout the agreement. If contract does not specify the date or delivery time, it should be delivered within a reasonable time after the contract is concluded. (Cruz-Mejia, et al., 2017) Besides, if the seller delivers before the time specified in the contract or delays in delivery, the buyer has the right to claim damages and reject the goods. Management of timeframe may have been a strategic significance, because it is a reflection of efficiency, expense, creativity and productivity management. Through that unnecessary time, it immediately increases other output metrics in a variable system. (J, et al., 1995)

#### 2.1.1.2 Receiving and Delivering Time

While in the delivery and logistics time, the company shall formulate a clear process to make systematic plans for pre-shipment delivery, processing of customs documents, issuance of delivery orders, arrangement of inventory store and logistical support, post-shipment customs control and other attempt to maintain the flow of data flow smoothly which can timely handling of problems. (Li, 2014) Collecting and shipping is a storage system for inbound and outbound content traffic. Inbound containers are delivered by truck to the depot, discharged at the

collecting terminals and placed into warehouse. Deliveries are obtained from warehouses, packed or prepared then delivered to consumers by docks using buses, lorries, trucks or ships. (CHOONG-YEUN LIONG, 2009)

The time when the goods are loading and unloading is very important as it can affect the lead time to deliver the goods. Consumers now are more emphasize on shorter lead time even the price of the products is higher. Some of them are willing to pay more to get the products more quickly. (Noori-Daryan, et al., 2019) European countries have now launched a new modern city freight service concept car to reduce the loading time and unloading time of goods which called FURBOT vehicle. In the modern era, urban freight services must be compatible with future Internet (IoT) e-commerce, artificial intelligence robots, multi-transportation mode linkages and low-carbon green environmental protection requirements. (Muscolo, et al., 2015) Autopilot is suitable in connecting vehicles and can increase the flexibility and safety. In addition, it can be automatically assembled into several different transport fleets according to the needs of cargo transportation so that the optimized fleet management system will provide more efficient services. (Zhang, et al., 2019)

#### 2.1.1.3 Just-in-Time System

Just-in-time is often defined by saying the correct amount of goods and services required at around the right time. It was a system of supplying goods as close as possible to when they are needed. Just-in-time is based on the logic that nothing will be produced until it is needed. For instance, when a product is sold, the system will pull a replacement product from the last position in the system and this process continues throughout the system. This system strives to eliminate waste such as inefficient setup procedures and inventories. Just-in-time focus on all of the aspects included in the production system as well as human resources, supply, inventories and technology. To be able to support a Just-in-time system, the firms required certain capabilities for example, short production lead time, economical small batch production and flexible resources. Furthermore to the rationale for quality-added and resource management for adoption Just-in-time, there are many factors behind Just-in-time adoption throughout the business. Causes behind implementing Just-in-time include raise organizational performance, enhance productivity, raise consumer loyalty, strengthen organizational-worker ties and achieve a sustainable competitive edge. Apart from that, Just-in-time brings massive of advantages such as removal of all content handling tends to result in reductions in labour expenses as a consequence to Just-in-time, decrease of fixed-up expenses, decrease of work-in-progress stock, decrease of turnaround time, increase in performance of inbound goods, fewer documentation, a substantial decrease in outbound finished items or facilities and a decline in the volume of unfairness reported among workers. (Lai & Cheng, 2009)

Besides adopt common supply materials such as JIT (Just-in-Time) or VMI (Vendor Management Inventory) and deliver goods as quickly as possible, the company can also sign an agreement with the manufacturer or supplier to guarantee supply. Production preparations will be planned as early as possible after the production planning including content planning, job orders, manufacturing processes, installation and testing of machinery. To improve the pace of output planning, a business can set up one plan to share the mechanism and quick review process for planning of the output. (Belalia & Ghaiti, 2016)

### 2.1.2 Reliability

#### 2.1.2.1 Accuracy of Online Order Service

Studies in lately show parcel delivery have gotten increasingly imperative to clients. It is conceivable to deliver more cost-adequately and responsively however requires a mix of digitization and outsourcing, giving more prominent control and insurance. The first step understands your clients and what they anticipate. Discovering delivery performance can encourage devotion when progressed nicely and drive clients away when it doesn't meet desires. (Iberahim, et al., 2016) When it comes to what is significant in delivery, the keyword has been speed for quite a while. Quicker rises to better and concentrate have been first on everyday conveyance and now plenty of retailers and e-commerce organizations are starting to offer same-day delivery. In any case, if speed is significant in conveyance, reliability, and precision is the king. Reliability in delivery service can ensure that clients get what they have requested precisely when they think they will get it. (Gulc, 2017) The need for client base needs to carefully consider

before putting resources into either quick of exact delivery. Clients must be informed about surprising occasions during the delivery procedure.

#### 2.1.2.2 Availability and Security of Website

Reliability is characterized as the proportion of consistency in terms of execution and just as the dependability. This measure has a few elucidations. It could be characterized as the firm playing out the service directly without rehashes or the firm regarding its guarantees. The measures most normally related with this trademark are accuracy and timeliness. (Choy, et al., 2013) The reliability delivery business was representing the capacity of the web site to satisfy order effectively, delivery expeditiously and keeps individual data secure. Also, clients contended that "reliability measurement" has an immediate beneficial outcome on perceived service quality and their fulfillment on E-commerce business. So far as the postal distribution company is concerned, reliability can indeed be estimated as the period required for distribute the shipment in the product delivery system as well as how long the package arrived at ideal location or final stop. (Omar, et al., 2015)

#### 2.1.2.3 Accurate Delivery Service

Reliability is referred to as the capacity an organization to play out its service accurately with dependably as somewhat of a guarantee. In certain words, dependability may be characterized as

just the organizational capacity for deliver on the guarantee, for example, guarantees about delivery service, issue goals, pricing and service arrangement. Since clients will in general trust and do organizations with the organizations which ready to hold their guarantee. (H.M.G.Y.J.Hennayake, 2017) Reliability has a noteworthy impact on clients and it was an investigator of consumer loyalty. For example, if an organization had the option to stay faithful to their commitments, consumer loyalty would increase. Conclusions regarding the outcome from past investigations, reliability was a significant persuasive issue of consumer satisfaction. (Mahfooz, 2014)

Delivery reliability can be improved through forwarding monitoring and expand product characteristics. In forward observing, the association has to realize what is on the loading list first, this regularly must be returned into the procedure at the checkpoint if there are deviations. Check further ahead of time to see the feasibility of load prior on. At that point intercede at a prior stage to guarantee that the material gets to the client on schedule. Besides, expanding the product characteristic. It related to the material is currently just arranged dependent on length, width, tallness and weight however during packing it can become evident that it is a lot for the truck. For example, an aluminum tube. With another pointer like 'volume for weight' it can create the impression that a package occupies an excessive amount of room. (Woeppel, 2014)

### 2.1.3 Responsiveness

#### 2.1.3.1 Problem Solving

Responsiveness measurements depend on the quick reaction and capacity to offer reviews if it does clients obtain an issue from the organization. Responsiveness recognizes The desire on staff to deal with the problem and attention to the expectations of the client had been a significant factor throughout the efficiency of courier services. (Gulc, 2017) Problem-solving strongly affect consumer loyalty if the expert staff helped them to solve the issue. (Mahfooz, 2014) As such, responsiveness seems to be the organizational efficiency which tests the capacity of delivery service suppliers as far as time, price and efficiency with respect to their customers. (Alminnourliza Noordin, n.d.) This measurement underscores attentiveness and promptness in handling consumer difficult, requests, troubles and complaints. Clients are told based on the amount regarding duration they need to request on seeking support, responses to queries or response to problems. In several other terms, the consistency of provision of services may be increased by responsiveness. (Omara, et al., 2016) This can help to provide the greatest chance to correct any issues before they become an issue for the customer.

#### 2.1.3.2 Interaction between Employees and Customer

The capacity to expertly quickly rebound from poor discernment may offer a good impression of consistency. Responsiveness causes a fundamental effect towards the impression of electronic service efficiency has an influence on aggregate of consumer loyalty recognition. (FIRDOUS, 2017) Employees should be capable to service the clients and give quick support to their clients.

This is because clients would show a negative impression of value if employees keep the clients sitting tight and waiting for no acceptable explanation. (Fadilah Siali, 2018) Customer service starts and ends with the people you have behind the desk. The right people have to be hired so that the customers will be taken care of. Employees should be ready to help the customer. (SAN, et al., 2013) When customers have doubts can contact the employees at any time and the employees should be answered as fast as possible. When the two sides have better communicate and interact, the transaction is continued to advance.

Customers commonly compare products or services. Therefore, employees should know about the related products well and the process of the shipment so they can respond on every consumer inquiry or request. (SAN, et al., 2013) The information requested by the customer such as value, distribution date, cost, currency, number, billing process, wrapping process, shipping method, location, contact individual and personal information should be recorded in the database and documented for the purpose of follow-up. Good customer service often provides the correct products or services to the entire consumer. (Jeske, et al., 2015)

#### 2.1.3.3 Access to Information

Moreover, provide customers the real-time information. At the point when both manufacturing activity and logistics activity, become "data transparent" and "advanced". This means that related

operators and customers in any location can know what is happening during the operation execution. (Zhang, et al., 2010) By using Wireless sensor networks can achieve this goal, it was A set of autonomous networks developed on or around the basis of the current web network. Radio Frequency Identification (RFID) has become among the hinge applications for wireless detector networks. This will gather details easily and precisely using Radio Frequency Identification technologies to allow the details transmit data with the conventional Web. (Chetouane, 2015) The actual usage of the network of integrated sensors can turn the logistics information management into intelligence. The intelligence, area, tracking, checking and the board of remote sensor system is exemplified throughout the domain of transportation as a unified information handling just as the system-based infrastructure focuses. This utilizes Radio Frequency Identification Devices (RFIDs) to gather content details and dispatch data to the signal processing facility across the web. (Xing, 2017)

### 2.1.4 Assurance

#### 2.1.4.1 Quality of Personnel

Assurance demonstrates the capacity to provide certainty and confidence to overall clients. Organizations have to ensure that their workers had prepared and learned the undertaking. Staffs also need to be friendly in responding grievance, issues and reviews from client. Employees who do achieve the certainty and support of clients have become the most important inquiries an organization will find them extensively. (Ma, 2012) This was because when clients were upset and not satisfied with the customer delivery service gave by workers, they will in a general

change to others companies and would not come back to do business with the past organization. (Fadilah Siali, 2018)

A skilled employee or staff would ensure their clients be given nice, compassionate and quick reaction which is the alternatives for providing clients with confidence and security. (SAN, et al., 2013) Employees can demonstrate their capacity to pass on confidence and certainty through a strong degree of comprehension and pleasantness. Customer care in the delivery process is ordinarily via telephone or different methods for electronic communication. To ensure customer service achieve customer satisfaction, the individual answerable to performance management should characterize value capacities on how to support clients. The quality care given assembles a strong connection in both the specialist service supplier and the customer. (Baruah, et al., 2015)

One of the most significant parts of distribution standard is assurance. This might be demonstrated by either the realities whereby clients decide in managing individuals or maybe even organizations which may believe. For example, assurance is expected in ensuring consumer loyalty. (Phiri & Mcwabe, 2013) Expectations for employees who are reliably respectful with clients were likewise high. Customers' expectations surpassed their observations in this service attribute. The conduct of employees, their behavior and attitude are unmistakable to clients and influence the delivery service experience either positively or negatively. (Hoffman & Bateson,

2001) Assurance can be enhancing through the improvement of employee behavior, ambiance infrastructure, lower charges, accessibility and great communication. The builds, for example, customer service, ease to utilize self-service innovation constructively affect the fulfillment of clients and satisfaction. (Subashini & Gopalasamy, 2016)

### 2.1.5 Delivery cost

#### 2.1.5.1 Competitive Advantage with Lower Cost

Among the most significant components for consumer, satisfaction is cost, and numerous researches have examined the connection between cost and consumer satisfaction. Cost is the fundamental component buyer think about when estimating the value of products and services. (Fadilah Siali, 2018) Clients will in general make a comparison between costs with the value or prevalence of products and services they got. Price mindfulness among clients goes about as a vital part of clients' decision-making activities. Previous studies show that clients' mindfulness on an organization's cost and contrasted it with its rivals' cost to build promising purposes and faithfulness towards that organization. (Mohammed, et al., 2017) Unless the expected interest or consumer loyalty with items was greater compare to its expense, clients will be happier and more likely to buy goods. (Nasirabadi & Bokaei, 2013) The increase or decrease in price give a tremendous effect on consumer satisfaction and brand exchanging. (SALEH, et al., 2015)

Delivery expenses relate to costs acquired by the business in getting its items to the hands of the clients. Customers' purchase items from the retail shops or e-retailers have absolutely relieed on the comparison of the expense of the item and its accessibility. Some e-retailers deduct their delivery cost along with the focused cost though the long-distance deliveries include the separate delivery cost to the progressively costly items. (Kennedy & Kundu, 2018) Customer likes lower cost items and don't incline toward delivery cost as it enhances the cost twofold the prize of the items. In this manner, higher delivery expenses are related to decrease ordering rates and arrangements that penalize bigger orders lead to reduced order size. (Vasić, et al., 2019)

#### 2.1.5.2 Free Delivery

Due to innovation advancement, a decrease in the costs by maintaining a strategic distance from unwanted traveling improves scores with the expectation of free deliveries. Free transporting enables the customer to search simply on the web and enhance sales. (Petrescu, 2011) This gives more data and time to analyze comparative items utilizing key factors, choosing of the item that suits the customer spending plan, how soon the item gets delivered, method of payments and the value of the items, this comparison and data lead the customer to think on the item that doesn't build the expense of the item and browse for a free delivery item in online increments. (Kennedy & Kundu, 2018) Additionally, online buyers scanning for comparative or interchange items to prevent transportation expenses or some of the time cancels the orders when shipping charge increase and retailers outlet charges are important to the delivery cost of the online buy. Delivery

cost at some point enhances the low-cost item which impact client to purchase items in retail locations. (Meng, et al., 2015)

## 2.2 Customer Satisfaction

J.L. Giese & J.A. Cote [2000] examined 20 customer satisfaction definitions that were used over 30 years and identified three general components:

1. Satisfaction is a mental or cognitive reaction.

2. The answer involves a specific focus, such as goals, product or service, knowledge.

3. Result takes place at such a specific period (since collection, intake or on the basis of cumulative knowledge). (Cichosz, et al., 2017)

Just-In-Time Logistics (JIT-L) which is an up-to-date idea can be characterized as the utilization of Just-In-Time the executive's concept of four significant transportation facilities, covering order handling, customer support, transportation management, and stock management. Customer services are the capability of transportation services that fulfill customers throughout terms of the time, communication, efficiency as well as comfort. This is also easier for rivals to emulate than cost reductions or any other innovative tactics Tech support also deals with the development of

location and time of operation in seven rights (7Rs). (OZALP, et al., 2010) For example, delivery is provided to the correct item in the right amount and right condition to the correct client during the ideal time at the right spot with the right delivery cost. (Shapiro & Heskett, 1995, p. 6). (Bielecki & Galinska, 2017). Improve in the efficiency of consumer services means the fulfillment of customer expectation will be increased when the movement of products or services are well handled. Consumer support is related to the performance of transportation networks and takes an significant part in generating, improving and sustaining client engagement and fulfillment. The operation of the application shall cover those actions of the order that gather, verify, entry and transfer request data. For instances, looking the accomplishment including exactness, testing customer confidence, demanding admission into the PC framework, sales and marketing division credits sales rep, accounting division records business transactions, inventory division finds the closest storage facility to the client then tells the people to fill up the request and distribution team assembles for shipping. This represents the partnership amongst vendors and companies. The delivery of the request applies to the date in which the vendor accepts the request before correct place has been approved to complete the transaction. Firms may reduce the order processing period via introducing Just-In-Time-Logistic successfully. (OZALP, et al., 2010) Thus, it is assumed that the timeliness, reliability provides significant impact in consumer loyalty of distribution services.

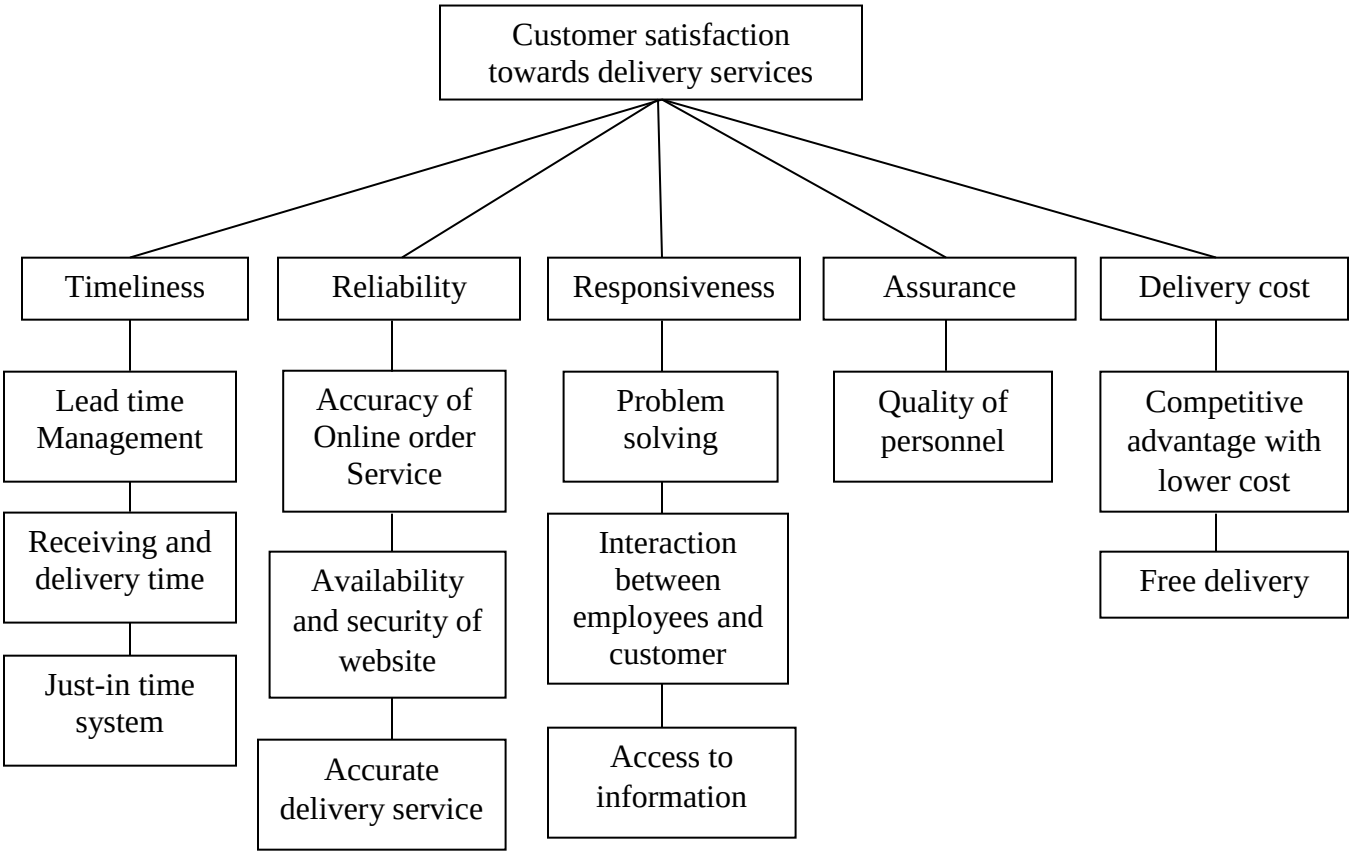
A customer is a person who creates value with the organization willingly effects on her / his behavior. Customer satisfaction is described as the emotions or attitude of the client towards a product or service by increasing revenue and profits through repeated purchases, the purchase of

fresh products and the buy of products by clients encouraged by satisfied clients. Customers who are highly satisfied with the organization offer other people favorable experiences and thus advertise for the organization, thereby reducing the price of attracting clients (Kavandi, Shakeri, 2010). (Gilaninia, et al., 2012) Higher profitability results from offering a big amount of after-sales service to customers (Rizaimy, Khaizir Muzani, Shamsul, & Wan, 2009). Current brand strategy reflects on the importance of a consumer's life as well as the preservation of deep-term client partnerships. From this point of view of the client relationship, from after-sales support is considered to be a major component that affect the establishment of healthy client relationships as well as deliver assurance to clients. On the other hand, in an extremely competitive market, the normal and perhaps optional generic after-sales support often plays such an significant part to gain further consumer interest. (Rizaimy, Khaizir Muzani, Shamsul, & Wan 2009). The delivery of suitable after-sales facilities to customers is a big source of employment, benefit and expertise in contemporary sectors. Acceptance of a decision which come back consumers become the highest efficient, requiring fewer selling work and creating relationships, after-sales support has an important role to play since it is a way of ensuring client satisfaction. (CHEE, 13April,2017) Thus, it is assumed that the assurance have significant effect in consumer loyalty with delivery services.

Consumer loyalty is based on variety of factors, including distribution charges, transit time, payment methods and information technology. Distribution is an important component of any logistic system. This involves the distribution of goods to the actual customer. Logistics processes consist of inputting, processing, shipping and delivering physical goods to final

consumers (Gunasekaran & Ngai, 2003). Nonetheless, the delivery of products incurs a reasonable fee which is usually paid by customers and which has an effect on customer satisfaction. The price of delivery (distribution charges) plays an important role in the degree of consumer service. Low delivery charges increase the level of satisfaction; nevertheless, high distribution charges reduce the level of satisfaction among consumers. In order to address this issue, new distribution channels can be introduced to increase customer satisfaction by lowering the distribution charges. It is also included in the refund the fee of the goods (Yuanxiao, 2014). Among the variables which effect consumer service, there is delivery costs linked to logistics. (Imran, et al., May 2018) Thus, it is assumed that the rate of distribution seems to have a significant impact in consumer engagement with transportation systems.

## 2.3 Purposed Theoretical Framework




The target from this analysis will be to examine influences that affect web customer contentment towards delivery services of Klang Valley. Based on the relevant theoretical framework that been stated in the previous section, there are five main criteria which are timeliness, reliability, responsiveness, assurance and delivery cost as well as sub-criteria are customer satisfaction towards distribution systems have been studied in this report.

# CHAPTER 3

# RESEARCH METHODOLOGY

## 3.0 Introduction

This section primarily explains in detail the study's research methodology, which involves research design, model development, sampling design, research instrument, data collection, and questionnaire design, as well as data analysis procedures.

## 3.1 Research Design

The research layout was a structure for study preparation and how analysis issues should be addressed. This requires the definition of analysis factors, the test methodology, test issues and data collection methods (Woodrow, 2014). The consistency and feasibility of the analysis rely on how the results, place, money and resources are gathered, calculated, evaluated and interpreted.

This analysis thesis is intended to analyze variables that impact online consumer satisfaction with Klang Valley products distribution channels using the Analytical Hierarchy Process (AHP) method. Which seeks to classify the ranking of five main criteria as well as the twelve sub-

criteria focused on previous user behavior with online transactions. As research studies the inclination of students for factors that can influence online consumer experience in the distribution of products services, this analysis is collected via the submission of questionnaires to Klang Valley students. There were a total of 15 places in the Klang Valley, namely Rawang, Klang, Serdang, Selayang, Shah Alam, Sepang, Gombak, Kajang, Putrajaya, Ampang, Puchong, Cyberjaya, W.P Kuala Lumpur ,Subang Jaya, Petaling Jaya. Both those students in Klang Valley may engage throughout the questionnaire as long as they meet the criteria. Among this research, 200 respondents among 18 and 25 years in age from Klang Valley were chosen as core respondents.

### 3.2 Data Collection

The primary data collection tool would be used in the whole survey review. Information collection performs a somewhat significant function in statistical research (Ajayi, 2017). In this research paper, the primary data collection approach is used to analyze variables that influence online consumer satisfaction with the distribution of product services throughout the Klang Valley.

#### 3.2.1 Primary Data

Predominant evidence was obtained with both the intention of seeking the answer to the issue through utilizing approaches such as surveys, interviews or tests. It relates to data produced for

the very first stage by researchers. Primary data was gathered throughout this thesis analysis through ego-administered survey forms. Ego-conducted questionnaire forms can be conducted online via the Online or by handing for each respondent in a printed form. (Ajayi, 2017)Feedback will be obtained earlier or updated once it has been done by all the respondents.

The system of questionnaires conducted through handing for each respondent was chosen to collect the primary data for this research report. The intention of handing distribution is that the sample group of this research project applies just to students who've had experience making purchases through online buying. The period of this study was roughly 3 weeks in which the questionnaires were circulated and obtained from selected respondents, between 1 Sep 2019 to 22 June 2019.

Furthermore, the explanation for the use of the hand-delivered method is that it is more accessible and reliable to the investigator while the scope of targeted respondents is limited. Interviewees must, therefore, be exempt from certain considerations and may honestly communicate their views. This questionnaire approach could only be studied on a limited scale. As somewhat of a consequence, researchers can retrieve further accurate knowledge.

### 3.3 Sampling Design

Sampling is the vital aspect of the result. Which helps the study to learn the connection between certain surveys as well as the community. That's because it's the method of picking a dataset to determine the features of the populace. In certain terms, that is the method of collecting knowledge about the whole community by analyzing just a portion of it (Kabir, 2016). Sampling which is often capable of delivering a variety of approaches to minimize the volume of data to be gathered and analyzed is known to be a post-group of the general population.

#### 3.3.1 Target Population

The population goal will be the local group on which information is sought while findings are taken from the analysis report (Kabir, 2016). The purpose of the analysis is to examine the factors influencing online consumer satisfaction with the distribution of services in the Klang Valley. Interviewees to this survey are also aimed at students who've had experiences purchasing items from an online retailer.

#### 3.3.2 Sampling Frame and Location

The sampling system is indeed a compilation of population components with correct contact details. It can be an address book, a community registry, a personnel list or a record of all

students attending a university (Kabir, 2016). The target demographic for this research analysis is 200 online shoppers in Klang Valley between the ages of 18 and 25.

### 3.3.3 Sampling Unit

Elementary elements or groups of these components that, besides being precisely specified, measurable and observable, are ideal for sampling purposes, are considered sampling units (Kabir, 2016). This research would concentrate on students in the Klang Valley.

### 3.3.4 Sampling Elements

Interviewees to the report contained online shoppers who had ordered and discounted products from an online retailer and were able to assess variables that have an impact on online overall satisfaction with retail distribution services throughout Klang Valley.

### 3.3.5 Sampling Size

The scale of the survey shall be the proportion of respondents throughout this research. The estimation of the sample size depends largely on the method of sampling system used (Kabir, 2016). Malaysian Communications as well as Multimedia Commission reported that the number

of online consumers in Malaysia rose from 76.9 percent in 2016 to 87.4 percent throughout 2018. Further than a quarter of online users using online transactions (54.2%) throughout 2018, which indicates that customers who engage in e-commerce is already gradually growing (Commission, 2018).

### 3.3.6 Sampling Method

Measurement methods can be categorized under two separate labels, which were the probability or non-probability of sampling. The sampling process specifies how and why the sample components were to be chosen. It can be influenced by both the study priorities, the allocation of financial capital, time limitations as well as the complexity of the issue to be examined (Kabir, 2016). The sampling method for non-probability was used in this analysis.

### 3.4 Model Development

The AHP template is structured towards address the Multiple Criteria Decision Making (MCDM) issue by decomposing the issue onto a structure. The structure comprises of several tiers, the peak, the center as well as the base. The highest point was the primary target, the middle stage is the judgment criterion, while the last point includes choices for judgment takes. The structure method is illustrated as follows:

**Figure 3.1: AHP index**

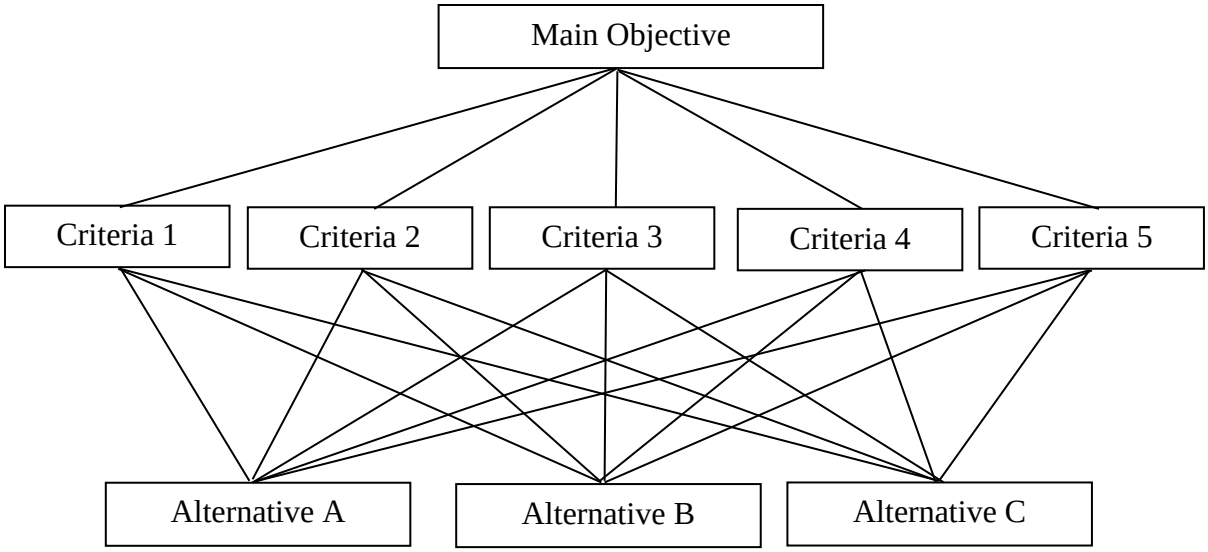



Table 3.1 highlights the hierarchy level throughout this analysis by pointing forth the key criterion for consumer satisfaction towards distribution services, the five judgment criteria as well as the twelve alternate decisions.

**Table 3.1: Hierarchy level of Consumer Satisfaction towards distribution services**

| | |
|---|---|
| **Top Level:** Main Objective | Consumer Satisfaction towards distribution services |
| **Second Level:** Main Criteria | 1. Timeliness<br>2. Reliability<br>3. Delivery cost<br>4. Responsiveness<br>5. Assurance |
| **Third Level:** Sub-Criteria | 1. Lead time Management<br>2. Receiving and delivery time<br>3. Just-in time system<br>4. Accuracy of Online order Service<br>5. Availability and security of website<br>6. Accurate delivery service<br>7. Problem solving<br>8. Interaction between employees and customer<br>9. Access to information<br>10. Quality of personnel<br>11. Competitive advantage with lower cost<br>12. Free delivery |

### 3.5 Research Instrument

#### 3.5.1 Questionnaires design

There are generally different information collection methods. The primary tools used in the research of blended methods are open-ended questionnaires, closed-ended questionnaires, surveys as well as classroom assessments. These distinct methods of collecting information that supplement one another, while increasing both data quality and reliability. Quantitative details are produced through closed-ended survey questions whereas qualitative knowledge is collected through open-ended survey questions. The survey elements are mainly drawn up as a result of research priorities and research problems (Zohrabi, February, 2013).

Results may be obtained through questionnaires or surveys that might be part of broader assessments around the same moment. The kind of problems we present in the questionnaires should help the respondents more relaxed, as in interviews as well as other equivalent tactics. We will, therefore, be conducted in such a un-intrusive way such that respondents might not get any feeling whereby their attitude, opinions regarding different dialects or semantic attitudes are assessed. Provided that concerns regarding language behaviors as well as how people describe oneself semantically are typically very sensitive topics, it is advised that they should be discussed implicitly or used in further detailed discussions. With regards to information concerning behaviors, there are all sorts of other records, along with age, level of education, family condition, nationality, state of residence, college enrollment and several other specifics that might be necessary for the study of results (Canals, 2017).

The key methods that we would like to utilize to obtain feedback are questionnaires. The questionnaire elements are primarily built onto the basis of study priorities or research issues. The questionnaire composed of several parts, including the predicted (A) population figures, (B) the main criteria for comparison and (C) the main criteria for comparison with the sub-criteria. The factors should be analyzed by the Analytic Hierarchy Process (AHP). The questionnaire is distributed by hand distribution using the non-probability sampling approach.

### 3.6 Data Analysis Procedures

The implementation of the AHP approach can be described in six easy steps (Atanasova–Pachemska, et al., 2014):

1. Classify the goal and discriminate between the decision criterion and the assessment strategy. This is creating a structured issue model that we shall make a choice regarding. The criterion (objective) was placed on top of the structure, the criteria including sub-criteria were set at both the reduced level as well as the alternatives were at the bottom of the scale. Figure 3.1 indicates the layout. (Atanasova–Pachemska, et al., 2014)

2. On each point of the structure, a distinction is rendered in pairs of structural components, where the priorities of the judgment-maker were articulated using a Saaty scale with comparative significance. The index comprises five rates and four sublevels that orally define the pressure, which related quantitative values throughout the scope 1 – 9 (Table 3.2). (Atanasova–Pachemska, et al., 2014)

**Table 3.2: Saaty Scale**

| **Numeric Rating / Significant** | **AHP Scale / Definition** | **Reciprocal (decimal)** |
|---|---|---|
| 1 | Equally Significant | 1/9 (0.111) |
| 2 | Equally to Moderate | 1/8 (0.125) |
| 3 | Moderate Significant | 1/7 (0.143) |
| 4 | Moderate to Strong | 1/6 (0.167) |
| 5 | Strong Significant | 1/5 (0.200) |
| 6 | Strong to Very Strong | 1/4 (0.250 |
| 7 | Very Strong Significant | 1/3 (0.333) |
| 8 | Very Strong to Extreme | 1/2 (0.500) |
| 9 | Extreme Significant | 1 (1.000) |

3. Generate a pairwise comparison array using the data collected.

   (a) When there are $\alpha$ of decision alternatives or $\alpha_1$ of decision criteria, there will be an ($\alpha$ x $\alpha$) pairwise comparison towards the analysis of decision criteria as well as a ($\alpha_1$ x $\alpha_1$) pairwise comparison regarding the evaluation of alternate decision-making in view of decision criteria. The array of reference will be built as follows:

$$C = \begin{bmatrix} C11 & \cdots & C1\alpha_1 \\ \vdots & \ddots & \vdots \\ 1/C1\alpha_1 & \cdots & C\alpha\alpha_1 \end{bmatrix}$$

   C $\alpha\alpha_1$ indicates the level of preference of component $\alpha$ to component $\alpha_1$.

   (b) When there are a variety of decision criteria nor alternate decisions, the number of pairwise contrasts would be specified as follows:

$$(0.5)\ \pi\ (\pi - 1)$$

4. The determination of the relative significance of the components at each point of the hierarchical system should be extended to the estimation of the main criterion, site-criteria nor alternatives. (Atanasova–Pachemska, et al., 2014) Measure the proportions for each and every decision criterion including alternate decisions using the Normalization Process. For this function, the amount of each column well into the matrix is calculated as well as all the components throughout the column were divided

by the entire column. The procedures are replicated with any of it pairwise contrast matrixes. For this study, five new normalized matrices are formed. Instead, the sum for each row throughout the newly formed matrices was the value or weight of the decision criteria as well as the alternate choice. The weight value from each decision criterion would be defined as $w^T$, whilst the weight score for all the decision alternatives of each decision criterion will be expressed as Q.

5. The total weight value of every alternate decision in array $\mathcal{F}$ is determined by multiplying Q with $w^T$. The equation is as follows:

$$\mathcal{F} = \mathrm{Q} \times \mathrm{w}^{\mathrm{T}}$$

The $\mathcal{F}$ array get to display the significance of alternate decisions. Components with both the greatest weight suggest the greatest or the highest-ranking of the alternative.

6. Test the accuracy in the pairwise matrix. Saaty (Table 3.4) suggested a consistency ratio (CR) which was determined in terms of the consistency index (CI) as well as the random index (RI) with the method below:

$$\mathrm{CR} = \frac{CI}{RI}$$

by this CI is determined as

$$\mathrm{CI} = \frac{\lambda\mathrm{max} - \mathrm{n}}{n - 1}$$

λmax is the supreme respective figure as well as n is the cumulative figure of parameters. λmax shall be calculated as below:

$$\frac{1}{n}\sum_{i=1}^{i=n} \frac{i^{\mathrm{th}}\ entry\ in\ A\mathrm{w}^{\mathrm{T}}}{i^{\mathrm{th}}\ entry\ in\ \mathrm{w}^{\mathrm{T}}}$$

Saaty (Table 3.2) produced RI by measuring the average CI of the completely random comparison matrix. In which the RI result is seized from Table 3.6 where the primary row ($n$) represents the figure of rows, such as the scale of the matrix, whilst the second row is the Random Consistency Index.

**Table 3.3 Random Consistency Index**

| $n$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| Random Index (RI) | 0 | 0.58 | 0.90 | 1.12 | 1.24 | 1.32 | 1.41 | 1.45 | 1.51 |

When CR ≤ 0.10, the degree of inconsistency in the pairwise contrast array is acceptable and afterward the outcome is reasonable and preferable since it indicates that the level of consistency is gratifying and hence the outcome is precise. On the contrary, the uncertainty is strong and the outcome-maker there is a need to re-evaluate the Cαα₁ of

matrix C to obtain greater consistency. When CR > 0.10, there could be major anomalies as well as the model does not yield a considerable outcome. (Atanasova–Pachemska, et al., 2014)

## 3.7 Conclusion

Methodologies used in this work, involving study design, data processing, sampling layout, model development, and test instrument as well as data analysis procedures. The study will analyze the data gathered from the participants via the questionnaire and perform the findings of the data research.

# CHAPTER 4

# RESULTS AND DISCUSSION

## 4.0 Introduction

This chapter would set out the opinion of online customers on the criteria for decisions, along with the alternatives of decisions. Each decision criterion together with decision alternatives are positioned to decide its weightage. The factors that can affect customer satisfaction towards goods delivery services in Klang Valley are ranked by concerning every main criterion. At last, all the weighting of each element is evaluated in order to identify it has the greatest effect on consumer experience. Advice and feedback are further set out in the last portion of this chapter.

## 4.1 Pairwise Comparison Matrices

The score of every criterion was determined with pair-wise comparison matrices. The choice was complete by comparing two alternatives against one criterion and the indicated priority will be shown after that. (Kie, et al., 2019) Segments at the more important stage of the hierarchy serve as a governing variable for segments at the lower rate. For instance, the top level of the hierarchy for this inquiry is the customer loyalty factor and the middle level is the five criteria. Therefore, The first pair-wise reference network is then a contrast with the five criteria. The tiar underneath is the third level which would be a sub-criteria. Only sub- of timeliness are contrasted under the

timeliness element. The same technique is used for the next four criteria. Components in the comparison frameworks are within the scale of 1-9. The diagonals in the frames are placed as "1" as they indicate the requirements or the option is compared with themselves. Due to there are five criteria, there is a sum of ten correlations for the principal grid. Investigation for this study is completed by using Microsoft Excel.

### 4.1.1 Comparison among Decision Criteria

As stated in Chapter 3, five of the decision criteria include timeliness, efficiency, sensitivity, and assurance and distribution costs. Table 4.1 presents the first 5 x 5 matrix for the determination of the five decision criteria.

**Table 4.1:** Pairwise comparison matrix for all five decision criteria

| Criteria | Timeliness | Reliability | Responsiveness | Assurance | Delivery Cost |
|---|---|---|---|---|---|
| Timeliness | 1.0000 | 0.5971 | 0.5236 | 0.5115 | 0.5732 |
| Reliability | 1.6748 | 1.0000 | 1.4110 | 1.1171 | 1.3027 |
| Responsiveness | 1.9099 | 0.7087 | 1.0000 | 1.0388 | 0.9637 |
| Assurance | 1.9549 | 0.8952 | 0.9626 | 1.0000 | 1.0729 |
| Delivery Cost | 1.7447 | 0.7676 | 1.0377 | 0.9320 | 1.0000 |
| Total | 8.2843 | 3.9686 | 4.9349 | 4.5995 | 4.9125 |

From Table 4.1 above, the inclining consistently stays 1.0000 as the decision criteria are contrasting with themselves. By pointing to the second row of the system, stability is more pronounced than the other four parameters. For instance, reliability is 1.6748 times greater important to timeliness, 1.4110 times more important than responsiveness and so on. However, for the worth which is underneath 1.0000, for example, 0.5115 under the comparison of timeliness and assurance, it implies that timeliness is a less important contrast with assurance. In other statements, assurance is 1.9549 more important than timeliness.

The recently framed normalized matrix is as appeared below since the grid within Table 4.1 is normalized to achieve the weight value of every decision criteria:

**Table 4.2:** Normalization of five decision criteria

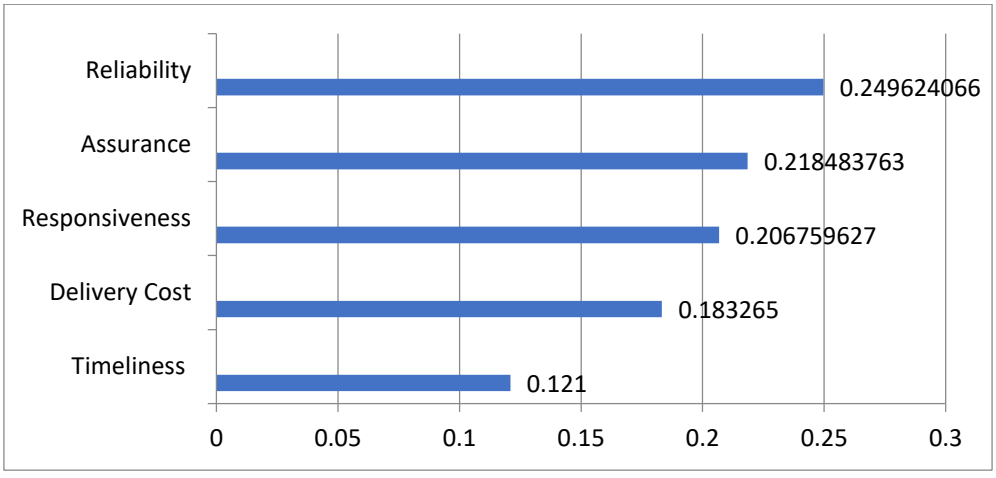
| Criteria | Timeliness | Reliability | Responsiveness | Assurance | Delivery Cost |
|---|---|---|---|---|---|
| Timeliness | 0.1207 | 0.150452902 | 0.106096977 | 0.111215293 | 0.116678749 |
| Reliability | 0.202164849 | 0.251976923 | 0.285921402 | 0.242876308 | 0.265180847 |
| Responsiveness | 0.230550568 | 0.178582075 | 0.202639339 | 0.225858599 | 0.196167555 |
| Assurance | 0.235975405 | 0.225559652 | 0.195062116 | 0.217413146 | 0.218408495 |
| Delivery Cost | 0.210598567 | 0.193428447 | 0.106096977 | 0.202636654 | 0.203564353 |

All normalized matrices would have an aggregate of 1.000 below the summation from each component in a similar column. The average value of each column is determined to examine the priority of these five decision criteria. The weight as well as ranking of five parameters are seen in Table 4.3 and Figure 4.1.

**Table 4.3:** Weight score and ranking of five decision criteria

| Criteria | Weightage | Rank |
|---|---|---|
| Timeliness | 0.1210 | 5 |
| Reliability | 0.249624066 | 1 |
| Responsiveness | 0.206759627 | 3 |
| Assurance | 0.218483763 | 2 |
| Delivery Cost | 0.183265 | 4 |

**Figure 4.1:** Weight of five decision criteria



As shown in Table 4.2 and the bar chart (Figure 4.1) above, reliability has the highest weight of 0.249624066 among the five decision parameters, which are the key factors that that influence the satisfaction of online customers in the distribution of products in Klang Valley. The second highest weight factor is an assurance that's 0.0313 smaller than reliability. The third and fourth considerations are responsiveness as well as delivery cost with a value of 0.206759627 and 0.183265 while timeliness which weighted 0.1210 ranked last.

### 4.1.2 Comparison among sub-criteria- Timeliness

The comparison matrix among sub-criteria of timeliness has shown in Table 4.4 below. It also shows consumers' preference reason when it come to the timeliness factor.

**Table 4.4** Comparison among sub-criteria- Timeliness

| Timeliness | Lead time management | Receiving and delivery time | Just-in time system |
|---|---|---|---|
| Lead time management | 1.0000 | 0.5344 | 0.7489 |
| Receiving and delivery time | 1.8713 | 1.0000 | 1.3355 |
| Just-in time system | 1.3353 | 0.7488 | 1.0000 |
| Total | 4.2066 | 2.2832 | 3.0844 |

The value of 1.8713, 1.0000 and 1.3355 show that receiving and delivery time of the goods is more preferred compare to the other two factors which are lead time management and just-in-time system in timeliness. Just-in-time system is also 1.3353 times more important than lead time management.

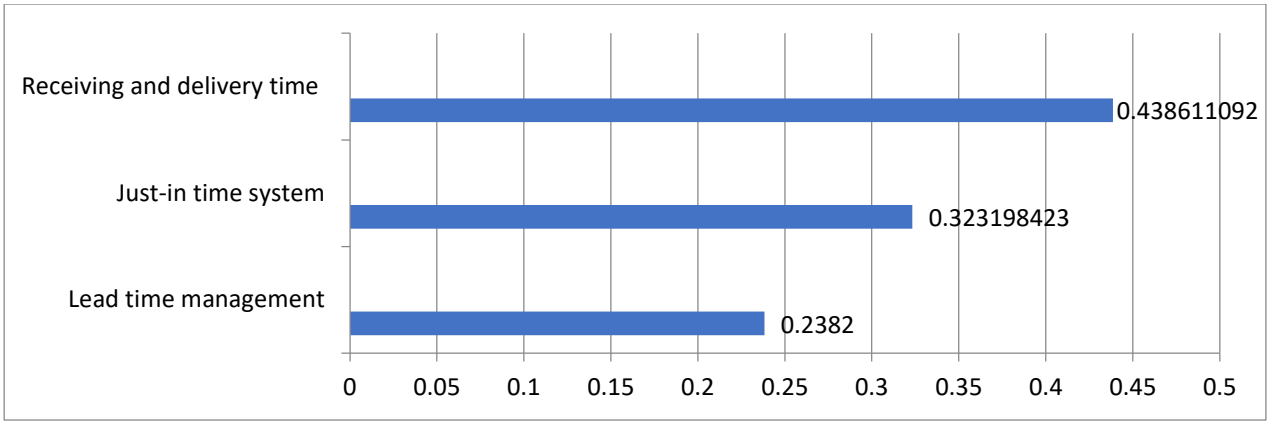
**Table 4.5:** Normalization of sub-criteria- Timeliness

| Timeliness | Lead time management | Receiving and delivery time | Just-in time system |
|---|---|---|---|
| Lead time management | 0.2377 | 0.234051662 | 0.242800514 |
| Receiving and delivery time | 0.444854284 | 0.437990917 | 0.432988076 |
| Just-in time system | 0.317426438 | 0.327957421 | 0.32421141 |

**Table 4.6:** Weight score and ranking of sub-criteria- Timeliness

| Timeliness | Weightage | Rank |
|---|---|---|
| Lead time management | 0.2382 | 3 |
| Receiving and delivery time | 0.438611092 | 1 |
| Just-in time system | 0.323198423 | 2 |

**Figure 4.2:** Results on timeliness factor




From Figure 4.2, the weight of receiving and delivery time which is 0.438611092 is the highest compare to the other two factors. Just-in-time system which weighted 0.323198423 is the second-highest and lastly is lead time management with 0.2382 of weight.

### 4.1.3 Comparison among sub-criteria- Reliability

The table below (Table 4.7) shows the matrix under the factor of reliability. It also shows consumers' preference reason in the reliability factor.

**Table 4.7** Comparison among sub-criteria- Reliability

| Reliability | Accuracy of online order service | Availability and security of website | Accurate delivery service |
|---|---|---|---|
| Accuracy of online order service | 1.0000 | 1.0440 | 0.9121 |
| Availability and security of website | 0.9579 | 1.0000 | 0.9663 |
| Accurate delivery service | 1.0964 | 1.0349 | 1.0000 |
| Total | 3.0543 | 3.0788 | 2.8784 |

The matrix in Table 4.7 shows the value of 1.0964, 1.0349 and 1.0000 in the factor of accurate delivery service is more preferred to the other factors. Because the quality of the electronic order process is 1.0440 times more critical than the availability and reliability of the website.

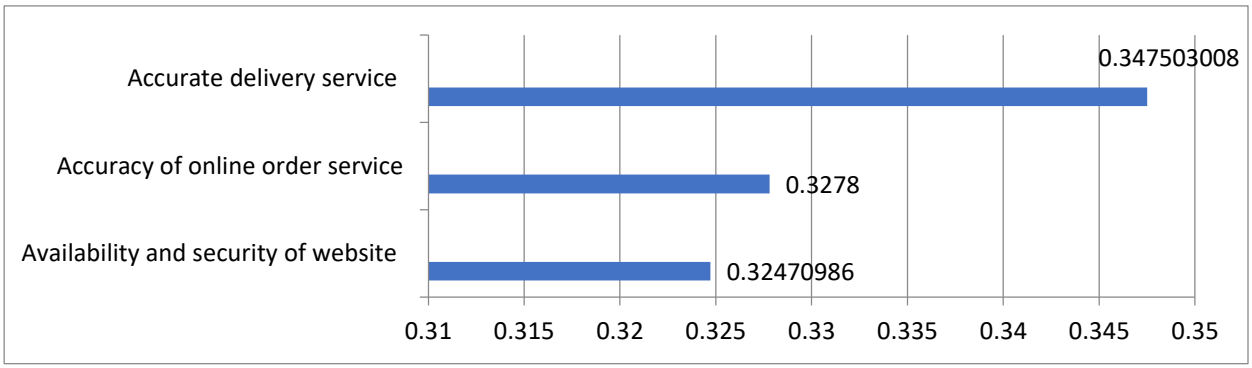
**Table 4.8:** Normalization of sub-criteria- Reliability

| Reliability | Accuracy of online order service | Availability and security of website | Accurate delivery service |
|---|---|---|---|
| Accuracy of online order service | 0.3274 | 0.339078562 | 0.316872893 |
| Availability and security of website | 0.313620874 | 0.324798063 | 0.335710643 |
| Accurate delivery service | 0.358969185 | 0.336123375 | 0.347416464 |

**Table 4.9:** Weight score and ranking of sub-criteria- Reliability

| Reliability | Weightage | Rank |
|---|---|---|
| Accuracy of online order service | 0.3278 | 2 |
| Availability and security of website | 0.32470986 | 3 |
| Accurate delivery service | 0.347503008 | 1 |

**Figure 4.3:** Results on reliability factor




Accurate delivery service ranked the highest with a weight of 0.347503008. The factor that ranked second is the accuracy of online order service showing weight with 0.3278. There is only 0.003 of different weight between the accuracy of online order service and availability and security of the website. It implies that the last two variables do not display a significant difference in their weight.

### 4.1.4 Comparison among sub-criteria- Responsiveness

Table 4.10 below outlines how the consumers rank the sub-criteria in the factor of responsiveness which included problem-solving, the interaction between employees and customers and access to information.

**Table 4.10** Comparison among sub-criteria- Responsiveness

| Responsiveness | Problem solving | Interaction between employees and customer | Access to information |
|---|---|---|---|
| Problem solving | 1.0000 | 1.6336 | 1.1633 |
| Interaction between employees and customer | 0.6121 | 1.0000 | 0.8866 |
| Access to information | 0.8596 | 1.1279 | 1.0000 |
| Total | 2.4717 | 3.7616 | 3.0499 |

Problem-solving is more preferable among other sub-criteria in responsiveness factor. Table 4.10 above shows that problem solving is 1.6336 times more important than the interaction between employees and customers and 1.1633 times more important than access to information. Besides, access to information is 1.1279 times more important than the interaction between employees and customers.

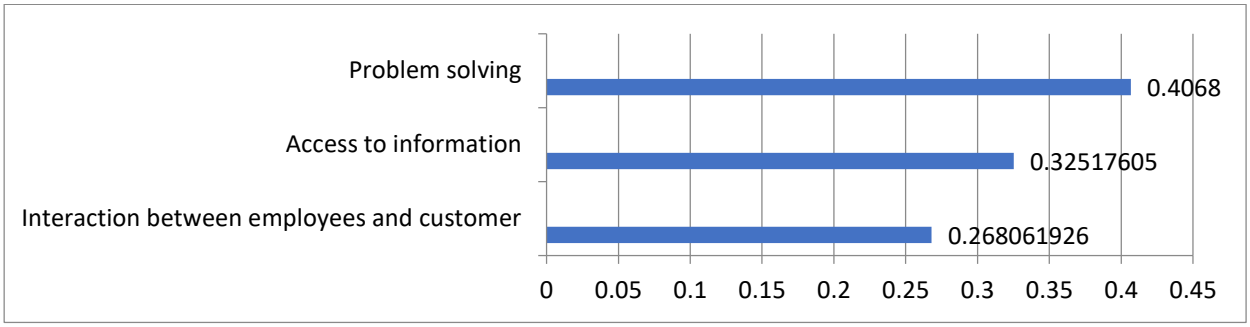
**Table 4.11:** Normalization of sub-criteria- Responsiveness

| Responsiveness | Problem solving | Interaction between employees and customer | Access to information |
|---|---|---|---|
| Problem solving | 0.4046 | 0.4342966 | 0.381431315 |
| Interaction between employees and customer | 0.247651626 | 0.26584538 | 0.290688772 |
| Access to information | 0.347773882 | 0.29985802 | 0.327879913 |

**Table 4.12:** Weight score and ranking of sub criteria- Responsiveness

| Responsiveness | Weightage | Rank |
|---|---|---|
| Problem solving | 0.4068 | 1 |
| Interaction between employees and customer | 0.268061926 | 3 |
| Access to information | 0.325170605 | 2 |

**Figure 4.4:** Results on responsiveness factor



From Figure 4.4, problem solving with a weight of 0.4068 tends to be the first goal preceded by exposure to knowledge with a weight of 0.325170605 and contact between workers and consumers with a weight of 0.268061926.

### 4.1.5 Comparison among sub-criteria- Assurance

There is no comparison among sub-criteria of assurance factor as it only has one sub-criteria which is the quality of personnel. When there is only one sub-criterion, the comparison is unable to perform. Therefore, normalization, weight score and ranking are also unable to calculate.

### 4.1.6 Comparison among sub-criteria- Delivery Cost

Delivery cost is also a comparable issue for the online consumer. Table 4.13 below shows the comparison of each sub-criteria in the factor of delivery cost.

**Table 4.13** Comparison among sub-criteria- Delivery Cost

| Delivery Cost | Competitive advantage with lower cost | Free delivery |
|---|---|---|
| Competitive advantage with lower cost | 1.0000 | 0.6818 |
| Free delivery | 1.4666 | 1.0000 |
| Total | 2.4666 | 1.6818 |

From the figures in Table 4.13, the inclining consistently stays 1.0000 as the sub-criteria are comparing with themselves. The value obtained from free delivery is of stronger importance compared to another sub-criterion. It is 1.4666 times more important than a competitive advantage with a lower cost.

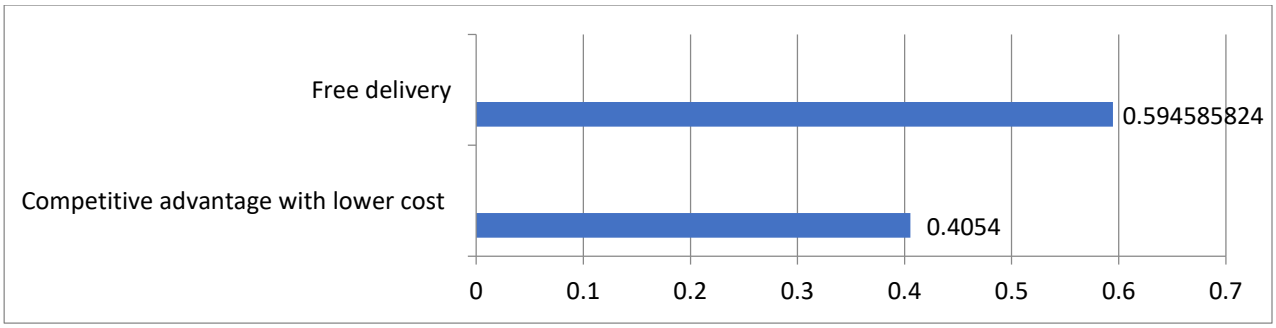
**Table 4.14:** Normalization of sub-criteria- Delivery Cost

| Delivery Cost | Competitive advantage with lower cost | Free delivery |
|---|---|---|
| Competitive advantage with lower cost | 0.4054 | 0.405414176 |
| Free delivery | 0.594585824 | 0.594585824 |

**Table 4.15:** Weight score and ranking of sub-criteria- Delivery Cost

| Delivery Cost | Weightage | Rank |
|---|---|---|
| Competitive advantage with lower cost | 0.4054 | 2 |
| Free delivery | 0.594585824 | 1 |

**Figure 4.5:** Results on delivery cost factor



There is 0.189 difference weight between free delivery and competitive advantage at lower cost. The weight of free delivery is 0.594585824. This means online consumers prefer free delivery compare to lower delivery costs.

## 4.2 Result Analysis

| Criteria | Weightage | Rank |
|---|---|---|
| Reliability | 0.249624066 | 1 |
| Assurance | 0.218483763 | 2 |
| Responsiveness | 0.206759627 | 3 |
| Delivery Cost | 0.183265 | 4 |
| Timeliness | 0.1210 | 5 |

In this study, reliability is proven as the main criterion that online consumers in Klang Valley will consider towards the satisfaction of goods delivery services. This is because most clients more focus on the reliability of delivery service when looking for products on the web, for example, whether the products are conveyed on schedule, with accuracy and a committed level of satisfaction. Digital companies will also be true to their commitments, bring the products to the right location and the right customers at the right moment. Besides, the guarantee of goods which got by the consumer is consistent with the information provided by the traders online can also affect consumer satisfaction. (Hong, et al., 2019)

Next, assurance ranked as second in the criteria. Numerous research have shown that there is a strong correlation between guarantee and service quality. Assurance is defined as the competence and politeness or willingness of workers to create confidence and loyalty that can have a direct effect on the degree of customer service. (Pakurár, et al., 2019)When the bridge of trust is built with clients, it is able to enhance their pleasant experience in online shopping. As a result, the satisfied customer is willing to come back to do business with the past seller again.

Responsiveness has the third-highest weight. This is due to responsiveness to clients is viewed as a significant indicator of service quality and driver of consumer satisfaction. Consumer responsiveness can positively be affecting consumer satisfaction and loyalty intervened by service performance on the point from view of the client. The outcome of service to the expectations of the consumers will lead to a positive assessment of the service procedure, henceforth a general fulfillment with the service exchange. (Uyoga, 2018) Although it might not the main reason that affect consumer satisfaction, firms used consumer responsiveness to improve the relationship with the consumer and upgrade all service to enhance effectiveness.

The last two decision criteria are the delivery cost and timeliness. The reason that delivery cost is fourth-ranking is that it does not affect much in consumer satisfaction. The delivery cost is cheaper when consumers buy goods in bulk, the more they buy the cheaper the cost. Moreover, sometimes the consumer is able to enjoy free shipping when they spend over a specific price.

Discounted shipping is also available when consumers consume a few times. At last, consumers select timeliness as the least considerable factor since this criterion is not a big problem for them. The estimate delivery time is stated before the purchase of the product. Therefore, the consumer is able to predict the time they are available to receive the goods. Consumers who buy products online instead of the physical stores are also willing to wait for the product to arrive in weeks.

### 4.3 Consistency Ratio

The consistency ratio is then determined based on the formula referenced in Chapter 3. To calculate the weighted sum vector (WSV), pairwise comparison matrix in Table 4.1 is used to multiply with weight score which is in Table 4.3.

$$WSV = \begin{bmatrix} 1.0000 & 0.5971 & 0.5236 & 0.5115 & 0.5732 \\ 1.6748 & 1.0000 & 1.4110 & 1.1171 & 1.3027 \\ 1.9099 & 0.7087 & 1.0000 & 1.0388 & 0.9637 \\ 1.9549 & 0.8952 & 0.9626 & 1.0000 & 1.0729 \\ 1.7447 & 0.7676 & 1.0377 & 0.9320 & 1.0000 \end{bmatrix} * \begin{bmatrix} 0.1210 \\ 0.249624066 \\ 0.206759627 \\ 0.218483763 \\ 0.183265 \end{bmatrix}$$

$$= \begin{bmatrix} 0.59513974696422 \\ 1.226869852 \\ 1.018413103 \\ 1.074196003 \\ 1.004234986 \end{bmatrix}$$

To calculate the consistency vector (CV), the components of the weighted sum vector are divided by the relative weights vector.

$$CV = \begin{bmatrix} 0.59513974696422 \\ 1.226869852 \\ 1.018413103 \\ 1.074196003 \\ 1.004234986 \end{bmatrix} \div \begin{bmatrix} 0.1210 \\ 0.249624066 \\ 0.206759627 \\ 0.218483763 \\ 0.183265 \end{bmatrix}$$

$$= \begin{bmatrix} 4.917254316 \\ 4.914870079 \\ 4.925589756 \\ 4.916594207 \\ 5.479687815 \end{bmatrix}$$

The pairwise analysis matrix is compatible if the own value, $\lambda_{max} \geq n$, where n denotes the figure of decision criteria.

$$\lambda_{max} = (4.917254316 + 4.914870079 + 4.925589756 + 4.916594207 + 5.479687815) / 5$$

$$= 5.030799235$$

$\lambda_{max}$ which is 5.030799235 is greater than 5.

$$CI = 5.030799235\text{-}5 / (5\text{-}1)$$

$$= 0.0077$$

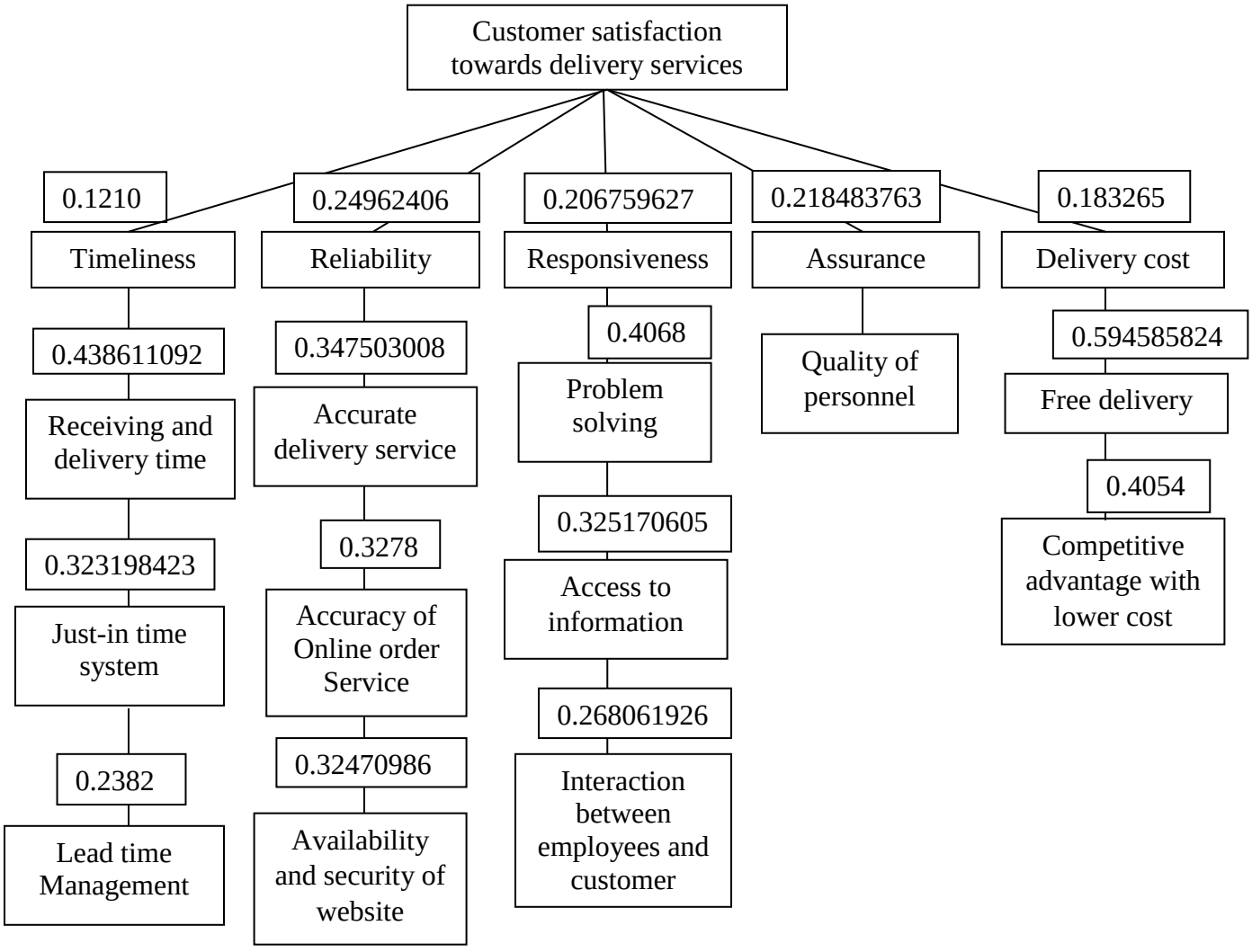
Since there are five parameters in this analysis, n=7 and RI=1.12 was used to determine the consistency ratio (CR).

$$CR = 0.0077 / 1.12$$

$$= 0.00687$$

The consistency ratio of 0.00687 < 0.10 has proved that the pairwise analysis indicated significant consistencies. After the investigation of the main criteria done, the research continued on the sub-criteria. The sub-criteria of each factor were paired against one another and positioned. The overall consistency ratio is also less than 0.10. Therefore, the outcome of this research using the Analytical Hierarchy Process (AHP) framework are satisfactory.

**4.4 Summary of Findings**



## 4.5 Recommendations

Although reliability is the most important factors that retain satisfaction of delivery services among online customer in Klang Valley, others factors such as assurance, responsiveness, delivery cost, and timeliness should also be improved to enhance the customer satisfaction. Even though timeliness has the lowest ranking, but it has to become better because online consumers are more preferred short delivery time instead of waiting a long time. Among the sub-criteria of reliability, accurate delivery service should be emphasized as it is more preferred by an online customers. The result also shows that problem solving is the most favorable among the sub-criteria of responsiveness. On the other hand of delivery costs, customers enjoy getting free delivery compared to competitive advantage at lower cost. Receiving and delivery time ranked as the most important sub-criteria in the timeliness factor. This result can be said that considering the essential needs of the consumer in the plan of delivery services and assume a significant job in hoisting the consumer satisfaction. It is recommended to think over the prioritization of these elements in this research project and take advantage of them in the advancement of related methodologies. In fact, the usage of the AHP model may enable researchers consider the highest weighting of decision-making parameters and discern the most appropriate choice based on the preference of decision-making alternatives. In conclusion, this calculation is able to help the option-makers to choose a most reasonable decision under a complex issue.

## 4.6 Contribution of the Research Project

An analysis of factors influencing consumer loyalty in the provision of services has been conducted by a variety of past scholars, but a very small amount of experiments have been carried out in Malaysia using the AHP model. The discoveries among this study can distinguish the most preferred factor that impacts online consumer satisfaction in Klang Valley thus it can be taken as a direction for the current delivery service provider to understand the need of consumers as well as to develop existing services.

# CHAPTER 5

# CONCLUSION

## 5.0 Introduction

Results from data collection and data interpretation in Chapter 4 will be covered in Chapter 5. The conclusion will come out with the research goals and study questions. In particular, the possible consequences of this study analysis will be described and addressed. The limitations of this project will also be concluded in this final chapter. Lastly, the last part will be covered by recommendations for further studies.

## 5.1 List of the Research Goals and Study Questions

The research results obtained from Chapter 4 which are also the contributions of this project have been highlighted in this section. The inference is based on research goals and study questions.

### 5.1.1 Research Questions One

Study Goals: To understand the factors that determine online consumer experience with delivery services.

Study Questions: What are the reasons that impact online consumer experience with delivery services?

The first goal and problem posed in this study is to evaluate the factors that impact online consumer experience with the delivery of services in the Klang Valley. According to most of the past researches done in a similar area, factors that can influence customer satisfaction towards delivery services included timeliness, reliability, responsiveness, assurance and delivery cost. These factors are the five main criteria of this research project. Some sub-criteria that can be affecting or get affected by them also have been examined through past studies. Under timeliness, there are sub-criteria such as lead time management, receiving and delivery time and just-in-time system. Sub-criteria that affect reliability factors are the accuracy of online order service, availability and security of the website and accurate delivery service. When coming up to determine how powerful the responsiveness factor is, sub-criteria including problem-solving, the interaction between employees and customers and access to information. The only sub-criteria that affect the assurance factor in customer satisfaction is the quality of personnel while sub-criteria that related to delivery cost have a competitive advantage with lower cost and free delivery. The extant literature of main criteria, sub-criteria and customer satisfaction have been discussed in Chapter 2.

### 5.1.2 Research Questions Two

Research Goals: To classify the ranking of factors influencing online customer satisfaction regarding distribution services by using the AHP model.

Research Questions: How online customers in Klang Valley ranked their satisfaction towards delivery service quality in terms of the factors?

Research question two has to identify how online customers evaluate the ranking of their satisfaction towards delivery service quality in terms of the main criteria. AHP model has been used in this research to answer this question. Both of the key criterion and sub-criteria has been contrasted to each other to analyze which has a stronger influence on consumer loyalty. Review of the findings in Chapter 4, the ranking of the main criteria is reliability has the highest priority among online customers, next is assurance, third is responsiveness, then delivery cost and last is timeliness. In terms of reliability criteria, customers feel that accurate delivery service is most important compare to the accuracy of online order service and availability and security of the website. While in the term of responsiveness, problem-solving has the highest rank, second is access to information and the last is the interaction between employees and customer. The result also showed that free delivery is more important than a competitive advantage with a lower cost. For the criteria of timeliness, we found out that receiving and delivery time has got the most priority of customers followed by just-in-time system and lead time management in timeliness criteria.

### 5.1.3 Research Questions Three

Study Goals: To ascertain the most important considerations for decision-making focused on online customers in the Klang Valley.

Research Questions: Which factor of delivery services is more preferred for online customers?

After ranking the main criteria using the AHP model, the results had shown that reliability is the most preferable factor among other main criteria for online customers in Klang Valley. This result also means that reliability meets the highest customers' priorities when it comes to online customer satisfaction towards delivery services. On the other hand, this research also discussed how important customer satisfaction towards goods delivery services quality in generating, improving and sustaining client engagement and also increasing organization profits through repeated purchases. Online customers will first consider the reliability factor of delivery services provider because it has the biggest impact on customer loyalty, accompanied by assurance, responsiveness, distribution cost and timeliness.

## 5.2 Implications for Theory

This research offered a deeper insight of the influencing factors online client satisfaction with the delivery of services in Klang Valley by discussed the most priority criteria that customers are involved in. Besides that, consumer purchasing behavior and decision are also related to this study. This is because this research had shown the main consideration of consumers when they want to purchase things online included select delivery services providers.

### 5.2.1 Implications for Delivery Services Providers

By highlighting the most preferable factors, delivery services providers are able to know which areas they should focus on their high performance. They get to know which criteria to highlight for the online consumer when they provide delivery services for them. This research can help the logistics and shipping industry to make sure which is the most important consideration when it comes to deliver goods as well as to achieve high customer satisfaction in order to get more market share of the overall industry. Delivery service providers can also realize how to gain competitive advantages to stand out among competitors. According to this research, the outcome proposed that they enhance the consistency of their distribution services in terms of timeliness, continuity, sensitivity, compliance and production costs. This is due to these factors had an important relationship towards customer satisfaction and had a direct effect on the customer buying decisions. Delivery service providers in Malaysia will concentrate more on content delivery, since it's the most crucial factor impacting internet consumer satisfaction. Moreover, the online consumers also cares about the assurance quality therefore the quality of personnel is

essential to deliver certainty and trust to their clients. Responsiveness also plays an important factor in order to solve the problem well, develop a good relationship and provide real-time information for online customer. In comparison, the rate of shipping could not be ignored as one of the key factors impacting consumer loyalty with the supply of products in Klang Valley. Even though timeliness is the least important compare to other factors, delivery services providers still need to stay alert that the goods always arrive on punctual. It is really necessary to insure that distribution services offered will fulfill the standards of online shoppers.

### 5.3 Limitation of Study

The limitation of this study has been identified to guarantee improvement in the future studies. Most of the respondents of this research project were 18-25 years old, therefore, the sample size collected from the respondents could not be indicative of the entire population in Klang Valley. This may causes the result analysis obtained here will in general be a predisposition to a fixed age group.

## 5.4 Future Research of Study

There are a range of guidelines and proposals for potential study. Extending the number of respondents to all age groups of the general population is recommended of order to improve the potential research. This research pays more attention to the young adults' point of view only. Although the sample size that selected is more representatives for the population that always purchases things online, teenagers who under 18 years old and housewives nowadays enjoy online purchasing as well as young adults. The point of view from the individuals in various age levels is able to obtain more reliable data about the most preferred main criteria and sub-criteria since online customers come from various age groups. In addition, the future study can conclude different main criteria and sub-criteria besides these criteria involved in this study. More and different decision criteria should be investigated to observe different results in significant criteria.

## 5.5 Conclusion

This study had achieved the objectives which are to recognize the priority of factors that affect user satisfaction with courier services online in Klang Valley. This research project also concludes that all factors had an important effect and direct relationship to online customer satisfaction towards delivery services. The main criteria obtained from this study are timeliness, reliability, responsiveness, assurance and delivery cost. In conclusion, the outcome of this study showed that reliability which had the highest priority is the top consideration followed by assurance, responsiveness, delivery cost and timeliness. Although reliability was the upmost essential component that impacts online customer happiness towards delivery services, but other

factors should not be underrated. Sub criteria of each main criteria have also been ranked in this research.

Moreover, the Analytical Hierarchy Process (AHP) model helps researchers to identify most proper decision under a complicated issue. The highest weightage of main criteria and sub-criteria had been determined through AHP calculation. Therefore, researchers are able to know the potential upgrades dependent on the most influential criteria. In addition, the consequences, drawbacks and suggestions for future study have been given in this project as a guidance for distribution service providers to establish a more fulfilled approach and comparison to consumer satisfaction for future research.

## References


Ajayi, O. V., 2017. Primary Sources of Data and Secondary Sources of Data. pp. 1-5.

Alminnourliza Noordin, N. H. a. N. H. O., n.d. SIX MAIN INNOVATION ISSUES: A CASE OF SERVICE INNOVATION OF POSTAL AND COURIER SERVICES IN MALAYSIA. pp. 1-10.

Apuke, O. D., 2017. QUANTITATIVE RESEARCH METHODS A SYNOPSIS APPROACH. *Arabian Journal of Business and Management Review (Kuwait Chapter),* pp. 40-47.

Atanasova–Pachemska, T., Lapevski, M. & Timovsk, R., 2014. ANALYTICAL HIERARCHICAL PROCESS (AHP) METHOD APPLICATION IN THE PROCESS OF SELECTION AND EVALUATION. *International Scientific Conference"UNITECH 2014" – Gabrovo ,* pp. 1-9.

Baruah, D., Nath, T. & Bora, D., 2015. Impact of Service Quality Dimensions on Customer Satisfaction in Telecom Sector. *International Journal of Engineering Trends and Technology (IJETT) – Volume 27 Number 2- September 2015,* pp. 1-7.

Belalia, Z. & Ghaiti, F., 2016. The value of Vendor Managed Inventory in an autocorrelated demand environment. *IFAC-PapersOnLine,* pp. 668-673.

Bielecki, M. & Galinska, B., 2017. TOTAL LOGISTICS MANAGEMENT CONCEPT AND PRINCIPLES IN MANUFACTURING ENTERPRISE. *17th international scientific conference Business Logistic in Modern Management October 12-13, 2017 - Osijek, Croatia,* pp. 1-15.

Bishop Emmanuel Prince, K. A. B., 2016. A Study on The Need to Implement a Courier Service Application on Android Smartphones. *nternational Journal of Information System and Engineering,* pp. 1-42.

Canals, L., 2017. Instruments for gathering data. *© 2017 Laia Canals (CC BY),* pp. 1-13.

CHEE, S. L. S., 13April,2017. THE EFFECT OF SUPPLY CHAIN MANAGEMENT ON CUSTOMER RETENTION AMONG GENERAL PRACTITIONERS IN PERAK: A CASE-STUDY OF INTERNATIONAL PHARMACEUTICAL COMPANY. *A research project submitted in partial fulfillment of the requirement for the degree of MASTER OF BUSINESS ADMINISTRATION (CORPORATE MANAGEMENT),* pp. 1-109.

Chetouane, F., 2015. An Overview on RFID Technology Instruction and Application. *IFAC-PapersOnLine,* pp. 382-387.

Chhetri, P. et al., 2017. Improving service responsiveness and delivery efficiency of retail networks. *International Journal of Retail & Distribution Management.*

CHOONG-YEUN LIONG, C. S. L., 2009. A SIMULATION STUDY OF WAREHOUSE LOADING AND UNLOADING SYSTEMS USING ARENA. *Journal of Quality Measurement and Analysis ,* pp. 1-12.

Chou, P.-F., 2014. AN EVALUATION OF SERVICE QUALITY, TRUST, AND CUSTOMER LOYALTY IN HOME-DELIVERY SERVICES. *International Journal of Research In Social Sciences ,* pp. 1-10.

Choy, M., Laik, M. N. & Shung, K. P., 2013. Performance Measurement Design for a Parcel Delivery Company. *Proceedings of the World Congress on Engineering 2013 Vol III,* pp. 1-6.

Cichosz, M., Goldsby, T. J., Knemeyer, A. & F.Tayor, D., 2017. INNOVATION IN LOGISTICS OUTSOURCING RELATIONSHIP – IN THE SEARCH OF CUSTOMER SATISFACTION. *2017, 13 (2), 209-219; p-ISSN 1895-2038 ; e-ISSN 1734-459X,* pp. 1-11.

Cihan, C. K. P. A. V., 2017. TRY NOT TO BE LATE! THE IMPORTANCE OF DELIVERY SERVICE IN ONLINE SHOPPING. *ORGANIZATIONS AND MARKETS IN EMERGING ECONOMIES, 2017, VOL. 8, No. 2(16),* pp. 177-181.

Commission, M. C. a. M., 2018. Internet Users Survey 2018. pp. 1-39.

Cruz-Mejia, O., Marmolejo, J. A. & Vasant, P., 2017. Lead time performance in a internet product delivery supply chain with automatic consolidation. *Journal of Ambient Intelligence and Humanized Computing,* pp. 1-12.

Fadilah Siali, A. W. S. W. &. M. U. A. H., 2018. Booming of Online Shopping in Malaysia: Do Customers Satisfy with Parcel Delivery Service?. *International Journal of Academic Research in Business and,* pp. 1-22.

Fall, C. 2., 2014. *Types of Data & Measurement Scales: Nominal, Ordinal, Interval and Ratio,* s.l.: s.n.

FIRDOUS, S., 2017. IMPACT OF INTERNET BANKING SERVICE QUALITY ON CUSTOMER SATISFACTION. *Journal of Internet Banking and Commerce, April 2017, vol. 22, no. 1,* pp. 1-17.

Gilaninia, S. et al., 2012. IMPACT OF SUPPLY CHAIN DIMENSIONS ON CUSTOMER SATISFACTION. *Kuwait Chapter of Arabian Journal of Business and Management Review Vol. 1, No.5; January 2012,* pp. 1-8.

Gulc, A., 2017. Courier service quality from the clients' perspective. *Engineering Management in Production and Services,* pp. 36-45.

Gulc, A., 2017. Models and Methods of Measuring the Quality of Logistic Service. *Procedia Engineering,* pp. 255-264.

H.M.G.Y.J.Hennayake, 2017. Impact of Service Quality on Customer Satisfaction of Public Sector Commercial Banks: A Study on Rural Economic Context. *International Journal of Scientific and Research Publications, Volume 7, Issue 2, February 2017,* pp. 1-6.

Hasim, M. S. et al., 2018. Commitment to sustainability: A content analysis of website for university organisations. *IOP Conf. Series: Earth and Environmental Science 117 (2018) 012046 doi :10.1088/1755-1315/117/1/012046,* pp. 1-9.

Hoffman, K. D. & Bateson, J. E., 2001. Essentials Of Services Marketing Hoffman. *2nd Edition, USA: South Western. Thomson Learning.,* pp. 1-9.

Hong, W., Zheng, C., Wu, L. & Pu, X., 2019. Analyzing the Relationship between Consumer Satisfaction and Fresh E-Commerce Logistics Service Using Text Mining Techniques. *Sustainability 2019, 11, 3570; doi:10.3390/su11133570,* pp. 1-16.

Iberahim, H., Mohd Taufik, N., Mohd Adzmir, A. & Saharuddin, H., 2016. Customer Satisfaction on Reliability and Responsiveness of Self Service Technology for Retail Banking Services. *Procedia Economics and Finance 37 ( 2016 ) 13 – 20 ,* pp. 13-20.

Imran, M., Hamida, S. N. b. A., Aziz, A. b. & Waseem-Ul-Hameed, May 2018. The contributing factors towards e-logistic customer satisfaction: a mediating role of information technology. *Uncertain Supply Chain Management 7 (2019) 63–72,* pp. 1-10.

Jacek Karcz, B. Ś., 2016. IMPROVEMENTS IN THE QUALITY OF COURIER DELIVERY. *International Journal for Quality Research 10(2) 355–372,* pp. 1-18.

Jeske, H., Chimusoro, E. & (PhD), A. M. K., 2015. AN EVALUATION OF CUSTOMER SERVICE AND THE IMPACT OFEFFICIENCY ON NAMIBIA'S LOGISTICAL SECTOR: A STUDY INVOLVING SELECTED COURIER COMPANIES. *SINGAPOREAN JOuRNAl Of buSINESS EcONOmIcS, ANd mANAGEmENt StudIES VOl.3, NO.6, 2015,* pp. 1-38.

Jessica Sze Yin Ho, D. O. L. T. F. T. L. F. K. T. Y. T., 2012. Logistic Service Quality among Courier Services in Malaysia. *2012 International Conference on Economics, Business Innovation IPEDR vol.38 (2012),* pp. 1-5.

J, R., Tersin & Hummingbird, E. A., 1995. Lead-time reduction: the search for competitive advantage. *International Journal of Operations & Production Management,* pp. 8-18.

Kabir, S. M. S., 2016. SAMPLE AND SAMPLING DESIGNS. pp. 168-172.

Kennedy, E. N. & Kundu, G. K., 2018. INFLUENCE OF DELIVERY CHARGES AND TIME ON ONLINE PURCHASE DECISION. *International Journal of Pure and Applied Mathematics,* pp. 1-12.

KEVIN CHAI YEE SHENG, L. Y. Y. L. W. L. O. S. P., 2018. FACTORS INFLUENCING CONSUMER SATISFACTION IN ONLINE SHOPPING. pp. 1-187.

Kie, C. J. et al., 2019. An Analytic Hierarchy Process Approach in Decision-Making for Material Selection in an Automotive Company: A Case Study. *FGIC 2nd Conference on Governance and Integrity 2019, KnE Social Sciences,* pp. 472-484.

Lai, K.-h. & Cheng, T., 2009. Just-in-Time Logistics. *Kee-hung Lai and T.C.E. Cheng 2009,* pp. 1-207.

Lee, C., 2001. Postal Services in the New Economy. pp. 1-16.

Li, X., 2014. Operations Management of Logistics and Supply Chain: Issues and Directions. *Discrete Dynamics in Nature and Society.*

Loeb, S. et al., 2017. Descriptive analysis in education:A guide for researchers. pp. 1-53.

Mahfooz, Y., 2014. Relationship between Service Quality and Customer Satisfaction in Hypermarkets of Saudi Arabia. *International Journal of Marketing Studies; Vol. 6, No. 4; 2014 ,* pp. 1-13.

Ma, Z., 2012. Assessing Serviceability and Reliability to Affect Customer Satisfaction of Internet Banking. *Journal of Software.*

Meng, Q.-c., Wan, X.-l. & Rong, X.-x., 2015. A Robust Ordering Strategy for Retailers Facing a Free Shipping Option. *PLoS ONE 10(5): e0125939..*

Mohammed, N. H. et al., 2017. Relationship among Service and Product Quality, and Price in Establishing Customer Satisfaction. *Journal of Biological and Environmental Sciences,* pp. 45-50.

Muscolo, G. G. et al., 2015. Industrial solutions for loading/unloading goods on a full electrical freight urban robotic vehicle. *Int. J. Vehicle Systems Modelling and Testing, Vol. 10, No. 4, 2015,* pp. 366-391.

Naser Valaei, S. R. M. K. S., 2016. CouQual: assessing overall service quality in courierservice industry and the moderating impact of age,gender and ethnicity. *Int. J. Management Concepts and Philosophy, Vol. 9, No. 2, 2016,* pp. 144-168.

Nasirabadi, E. Z. & Bokaei, M. N., 2013. A study on relationship of brand characteristics and customer satisfaction with brand loyalty. *Management Science Letters, 3(13), 1907-1914. ,* pp. 1907-1914.

Niels Agatz, A. M. C. M. F. M., n.d. Challenges and Opportunities in Attended Home Delivery. pp. 1-18.

Noori-Daryan, l., Taleizadeh, A. A. & Jolai, F., 2019. Analyzing pricing, promised delivery lead time, supplier-selection, and ordering decisions of a multi-national supply chain under uncertain environment. *International Journal of Production Economics,* pp. 236-248.

Nurul Izzah, D. R. L. Y., 2016. Relationship-Courier Partner Logistics and E-Commerce Enterprises in Malaysia: A Review. *Indian Journal of Science and Technology, Vol 9(9),* pp. 1-10.

Okyere, S., (Ph.D), J. A. & Anning, K. S., 2015. Evaluating the Effect of Lead Time on Quality Service Delivery in the Banking Industry in Kumasi Metropolis of Ghana. *Journal of Arts & Humanities,* pp. 1-16.

Omara, M. S., Ariffinb, H. F. & Ahmadc, R., 2016. Service Quality, Customers' Satisfaction and the Moderating Effects of Gender: A Study of Arabic Restaurants. *Procedia - Social and Behavioral Sciences 224 ( 2016 ) 384 – 392 ,* pp. 384-392.

Omar, H. F. H., Saadan, D. K. B. & Seman, P. K. B., 2015. Determining the Influence of the Reliability of Service Quality on Customer Satisfaction: The Case of Libyan E-Commerce Customers. *International Journal of Learning & Development,* pp. 1-4.

Otsetova, A., 2017. RELATIONSHIP BETWEEN LOGISTICS SERVICE QUALITY, CUSTOMER SATISFACTION AND LOYALTY IN COURIER SERVICES INDUSTRY. *MANAGEMENT AND EDUCATION TOM XIII (2) 2017 VOL. XIII (2) 2017,* pp. 51-57.

OZALP, I., SUVACI, B. & TONUS, H. Z., 2010. A NEW APPROACH IN LOGISTICS MANAGEMENT: JUST IN TIME-LOGISTICS (JIT-L). *INTERNATIONAL JOURNAL OF BUSINESS AND MANAGEMENT STUDIES Vol 2, No 1, 2010 ISSN: 1309-8047 (Online),* pp. 1-9.

Pakurár, M. et al., 2019. The Service Quality Dimensions that Affect Customer Satisfaction in the Jordanian Banking Sector. *Sustainability 2019, 11, 1113; doi:10.3390/su11041113,* pp. 1-24.

Petrescu, M., 2011. Online price dispersion – more than imperfect information. *Journal of Product & Brand Management 20(7):541-548,* pp. 541-548.

Phiri, D. M. A. & Mcwabe, T., 2013. CUSTOMERS' EXPECTATIONS AND PERCEPTIONS OF SERVICE QUALITY: THE CASE OF PICK N PAY SUPERMARKET STORES IN PIETERMARITZBURG AREA, SOUTH AFRICA. *International Journal of Research In Social Sciences ,* pp. 1-9.

Piruni A. Deyalage, D. K., 2019. Factors Affecting Online Customer Satisfaction: The Sri Lankan Perspective. *International Journal of Business and Management; Vol. 14, No. 2; 2019,* pp. 99-103.

S. Akilimalissiga, N. S. A. V., 2017. The Delivery of Service Quality to Increase Customer Repurchase Behaviour and Customer Satisfaction at Fast Food Outlets in Central Johannesburg, South Africa. pp. 1-6.

SALEH, M. A. H. et al., 2015. Customer Satisfaction and Brand Switching Intention: A Study of Mobile Services in Saudi Arabia. *Expert Journal of Marketing, 3(2), pp. 62-72,* pp. 62-72.

SAN, G. P. et al., 2013. THE RELATIONSHIP BETWEEN SERVICE QUALITY AND CUSTOMER SATISFACTION:AN EMPIRICAL STUDY IN BANKING INDUSTRY. pp. 1-132.

Saunders, M., Lewis, P. & Thornhill, A., 2009. Research Methods for Business Students. *Research Methods for Business Students fifth edition,* pp. 1-649.

Subashini, R. & Gopalasamy, V., 2016. A REVIEW OF SERVICE QUALITY AND CUSTOMER SATISFACTION IN BANKING SERVICES: GLOBAL SCENARIO. *The Journal of Internet Banking and Commerce.*

Usmana, M. & Rehman, Z. U., 2017. The Impact of Customer Satisfaction on Price Sensitivity in Courier Services Sector. *Journal Of Humanities And Social Science (IOSR-JHSS) ,* pp. 1-14.

Uyoga, D., 2018. Relationship between Customer Responsiveness, Service Performance and Satisfaction among Airline Passengers in Kenya. pp. 1-13.

Vasić, N., Kilibarda, M. & Kaurin, T., 2019. The Influence of Online Shopping Determinants on Customer Satisfaction in the Serbian Market. *Journal of Theoretical and Applied Electronic Commerce Research,* pp. 1-20.

Woeppel, M., 2014. HOW TO BUILD A RELIABLE DELIVERY PROCESS. pp. 1-11.

Woodrow, L., 2014. Writing about Research. *L. Woodrow, Writing about Quantitative Research in Applied Linguistics,* pp. 1-2.

Xing, J., 2017. An Intelligent Logistics Tracking System Based on Wireless Sensor Network. *Paper–An Intelligent Logistics Tracking System Based on Wireless Sensor Network,* pp. 1-12.

YUSSOFF, Y. B. M., 2014. CREATING SUSTAINABLE COMPETITIVE ADVANTAGE THROUGH NTAGE THROUGH A CASE STUDY OF POSLAJU IN POS MALAYSIA BERHAD. *RESEARCH REPORT IN PARTIAL FULFILMENT OF THE REQUIREMENTS,* pp. 1-34.

Zhang, P. G., Li, P. X. & Diabat, P. A., 2019. “Intelligent Transportation and Logistics in Manufacturing with Big Data”. *International Journal of Production Economics.*

Zhang, Y. et al., 2010. RFID-enabled real-time manufacturing information tracking infrastructure for extended enterprises. *J Intell Manuf (2012) 23:2357–2366 DOI 10.1007/s10845-010-0475-3,* pp. 1-10.

Zohrabi, M., February, 2013. Mixed Method Research: Instruments, Validity, Reliability and Reporting Findings. *Theory and Practice in Language Studies, Vol. 3, No. 2, pp. 254-262,,* pp. 1-9.

Zulkarnain Kedah, Y. I. A. A. H. &. S. A., 2015. Key Success Factors of Online Food Ordering Services: An Empirical Study. pp. 1-18.

**APPENDICES: QUESTIONNAIRES**

## Definition of 5 decision criteria

1. Timeliness – Timeliness is the time limit for the seller to deliver the contracted goods to the buyer or carrier as specified in the sales contract. Timeliness of a contract is deemed as a warranty.

2. Reliability - Reliability in delivery service can ensure that clients get what they have requested precisely when they think they will get it. It ensure that the order and delivery is accurate.

3. Responsiveness - Responsiveness measurements depend on the quick reaction and capacity to give feedback if the clients have issue to the organization. It recognizes the willingness of workers to solve an issue and their responsiveness towards the client needs.

4. Assurance - Assurance shows the capacity to deliver certainty and trust to their clients. The organization must ensure their workers were well preparing and learned about their undertaking.

5. Delivery cost - Cost is the fundamental component buyer think about when estimate the value of a products and services. Delivery expenses relate to costs acquired by the business in getting its items to the hands of the clients

# UDPL3246 RESERCH PROJECT

# Title: Factors that affect online customer satisfaction towards goods delivery services in Klang Valley.

We are Year 3 undergraduate student pursuing Bachelor of Science (Honours) Logistics and International Shipping who currently conducting a questionnaire on preference on the factors that affect online customer satisfaction towards goods delivery services in Klang Valley among students for our final year project. This questionnaire is designed to identify the priority of the factors that can affect online customer satisfaction in delivery services from the students' points of view. This questionnaire consists of 5 parts which is part A, B, C, D and E.

Age

☐ Below 18 ☐ 18-25 ☐ Above 25

Gender

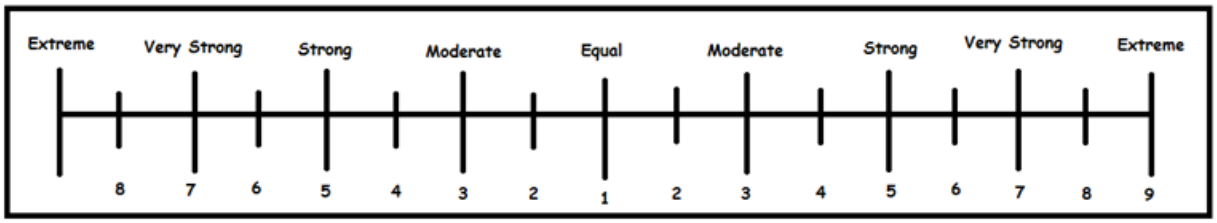
☐ Male ☐ Female

Area in Klang Valley

| | | | | |
|---|---|---|---|---|
| ☐ Rawang | ☐ Selayang | ☐ Gombak | ☐ Ampang | ☐ Kuala Lumpur |
| ☐ Klang | ☐ Shah Alam | ☐ Kajang | ☐ Puchong | ☐ Petaling Jaya |
| ☐ Serdang | ☐ Sepang | ☐ Putrajaya | ☐ Cyberjaya | ☐ Subang Jaya |

How often you purchase things online in a month (times) : _____

**SCALE:**




**METHOD OF ANSWERING:**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | (9) | 8 | 7 | 6 | 5 | 4 | 3 | 2 | **1** | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | B |

I prefer "A" 9 times more than "B"

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | **1** | 2 | 3 | 4 | 5 | (6) | 7 | 8 | 9 | B |

I prefer "B" 6 times more than "A"

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | (1) | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | B |

Both "A" and "B" are of equal importance to me.

**Part A**

**Criteria Level**

The following is a listing which may have influence on your **decision making.**

**Compare the relative IMPORTANCE with respect: FACTOR.**

i) Please **circle** only **ONE** number you **most agree** in each row

**(Scale 1 in middle box if both criteria have similar level of effect)**

| | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Timeliness | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Reliability |
| Timeliness | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Responsiveness |
| Timeliness | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Assurance |
| Timeliness | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Delivery cost |
| | | | | | | | | | | | | | | | | | | |
| Reliability | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Responsiveness |
| Reliability | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Assurance |
| Reliability | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Delivery cost |
| | | | | | | | | | | | | | | | | | | |
| Responsiveness | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Assurance |
| Responsiveness | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Delivery cost |
| | | | | | | | | | | | | | | | | | | |
| Assurance | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Delivery cost |
| | | | | | | | | | | | | | | | | | | |

**Part B:**

**Sub-criteria Level**

The following is a listing which may have influence on your **decision making.**

**Compare the relative IMPORTANCE with respect: SUB-CRITERIA OF TIMELINESS**

ii) Please **circle** only **ONE** number you **most agree** in each row

**(Scale 1 in middle box if both criteria have similar level of effect)**

| | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Lead time management | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Receiving and delivery time |
| Lead time management | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Just-in time system |
| | | | | | | | | | | | | | | | | | | |
| Receiving and delivery time | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Just-in time system |
| | | | | | | | | | | | | | | | | | | |

**Part C:**

**Sub-criteria Level**

The following is a listing which may have influence on your **decision making.**

**Compare the relative IMPORTANCE with respect: SUB-CRITERIA OF RELIABILITY**

iii) Please **circle** only **ONE** number you **most agree** in each row

**(Scale 1 in middle box if both criteria have similar level of effect)**

| | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Accuracy of online order service | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Availability and security of website |
| Accuracy of online order service | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Accurate delivery service |
| | | | | | | | | | | | | | | | | | | |
| Availability and security of website | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Accurate delivery service |
| | | | | | | | | | | | | | | | | | | |

**Part D:**

**Sub-criteria Level**

The following is a listing which may have influence on your **decision making.**

**Compare the relative IMPORTANCE with respect: SUB-CRITERIA OF RESPONSIVENESS**

iv) Please **circle** only **ONE** number you **most agree** in each row

**(Scale 1 in middle box if both criteria have similar level of effect)**

| | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Problem solving | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Interaction between employees and customer |
| Problem solving | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Access to information |
| | | | | | | | | | | | | | | | | | | |
| Interaction between employees and customer | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Access to information |
| | | | | | | | | | | | | | | | | | | |

**Part E:**

**Sub-criteria Level**

The following is a listing which may have influence on your **decision making.**

**Compare the relative IMPORTANCE with respect: SUB-CRITERIA OF DELIVERY COST**

v) Please **circle** only **ONE** number you **most agree** in each row

**(Scale 1 in middle box if both criteria have similar level of effect)**

| | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Competitive advantage with lower cost | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Free delivery |
| | | | | | | | | | | | | | | | | | | |